

Chance Constrained Optimization with Kernel Density Estimation

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1 The Problem of Optimization Under Uncertainty

Every optimization model is a promise. It promises that if the decision it recommends is implemented, the constraints of the system will hold and the objective will be as good as the mathematics claims. That promise is honest only if the numbers inside the model are honest, and in almost every real deployment they are not. The demand a grid operator schedules against is a forecast, not a fact. The thrust a lander commands is corrupted by disturbances that sensors resolve only imperfectly. The returns a portfolio is built on are realizations of a process that is not available in closed form. This chapter establishes precisely what goes wrong when uncertainty is ignored, why the two classical remedies, ignoring it and armoring against the worst case, each have important limitations in the settings this thesis targets, and why the chance constraint, a constraint required to hold with a prescribed probability, offers a principled middle ground between them. It then exposes the three structural reasons chance constraints are computationally hard, builds the case for estimating the relevant probability distributions from data rather than assuming them, and introduces the single conceptual shift on which the entire thesis rests: moving the estimation problem from the space of the uncertainty to the space of the constraint residual. The chapter closes by fixing the scope, the anchor literature, and the research questions of the thesis.

1.1 Where Deterministic Optimization Falls Short

This section establishes why the deterministic template struggles under uncertainty. It first identifies the mechanism behind the failure, then illustrates it across four application domains, works through a numerical example in which the failure can be quantified exactly, and closes by showing that the observed failure rate is largely independent of the application.

1.1.1 The mean-value approach and the flaw of averages

The standard nonlinear program

$$\underset{x \in X}{\text{minimize}} \ f(x) \quad \text{subject to} \quad g(x, \xi) \leq 0, \tag{1}$$

treats the parameter vector ξ as a known constant. Decades of solver development have made problems of the form Eq. (1) routinely solvable at scale, and this success

has had a quiet side effect: the deterministic template is applied by default, even when ξ is plainly a random quantity. The usual workaround is to replace ξ with a point estimate, most often its mean or a nominal forecast, and solve the resulting deterministic problem. This is the mean-value approach, and its shortcomings are systematic rather than occasional.

The failure mechanism is worth stating carefully, because it is the mechanism behind the examples in this section. An optimizer pushes its solution to the boundary of the feasible region; that is what optimality means whenever a constraint is binding. If the constraint $g(x, \bar{\xi}) \leq 0$ is active at the mean-value optimum x_{mv}^* , then the residual $g(x_{mv}^*, \xi)$ sits exactly at zero when ξ equals its mean. The moment ξ is allowed to fluctuate around that mean, roughly half of its realizations push the residual positive and the constraint is violated. The optimizer does not merely tolerate risk; it tends to select the configuration in which the risk of violation is largest, because that configuration is where the objective is best. In this sense optimization and uncertainty often work against each other, a phenomenon known in the planning literature as the flaw of averages (Savage, 2009): a plan built on average inputs need not behave like an average plan, and can be considerably more fragile.

1.1.2 Four domains where uncertainty bites

Four application domains make this concrete. They are chosen deliberately, because each reappears later in the thesis as a benchmark or a case study, and because together they span the distributional pathologies that motivate the nonparametric approach of Section 1.4.

Autonomous lunar landing. A powered-descent guidance problem asks for a thrust profile that brings a lander from orbit to a designated site while respecting bounds on velocity, glide slope, and fuel. The dynamics are integrated forward under thrust perturbations and imperfectly known gravitational and terrain parameters. Keil u. a. (2021) show that disturbance distributions in this setting are naturally non-Gaussian; a mixture arises, for example, when a thruster operates in one of two regimes. A trajectory optimized for nominal thrust grazes the velocity constraint at touchdown, so under perturbation the lander exceeds its safe impact speed on a large fraction of descents. A lunar landing offers no opportunity to repeat a failed attempt. Chance-constrained formulations of exactly this problem are the application anchor of the thesis (Keil u. a., 2020, 2021; Blackmore u. a., 2010).

Energy dispatch under renewable generation. A system operator commits generation a day ahead against uncertain demand and uncertain wind and solar output. Forecast errors for renewable infeed are well documented to be skewed and heavy-tailed rather than Gaussian (Hodge u. Milligan, 2011; Bienstock u. a., 2014). A dispatch that balances the forecast exactly is short of energy whenever net demand lands above its forecast, which by construction happens with probability near one half, and reserve activations, load shedding, or price spikes follow. This domain has produced some of the strongest data-driven chance-constraint literature (Ciftci u. a., 2019; Wu u. a., 2021; Wu u. Kargarian, 2023).

Chemical process control. A reactor recipe is optimized for yield subject to temperature and purity limits, but the composition of the feed material varies from batch to batch. Running the process at the deterministic optimum places the temperature constraint on the boundary, and ordinary feedstock variation then produces off-specification batches at a steady rate. Process systems engineering was, in fact, the field where data-driven kernel-smoothed chance constraints were first formulated (Calfa u. a., 2015).

Portfolio allocation. An investor maximizes expected return subject to a cap on the probability of a large loss. Asset returns exhibit heavy tails and volatility clustering; these are among the most robust stylized facts in empirical finance (Cont, 2001). A portfolio optimized as if returns were fixed at their historical means concentrates in the highest-mean assets and breaches its loss cap far more often than any Gaussian calculation predicts (Rockafellar u. Uryasev, 2000; Liu u. a., 2022).

1.1.3 A worked example: the cost of trusting the mean

The mechanism is easiest to see in a problem small enough to solve by inspection. Consider a single-bus dispatch model. A controllable resource of capacity x megawatts must cover an uncertain net demand ξ , and capacity is costly, so the planner solves

$$\underset{x \geq 0}{\text{minimize}} \quad c x \quad \text{subject to} \quad \xi - x \leq 0. \quad (2)$$

Let net demand follow a moderately skewed lognormal law with mean $\bar{\xi} \approx 102$ MW, a realistic shape for net load once renewable infeed is subtracted (Hodge u. Milligan, 2011). The mean-value method replaces ξ by $\bar{\xi}$ and returns $x_{\text{mv}}^* = \bar{\xi}$: build exactly

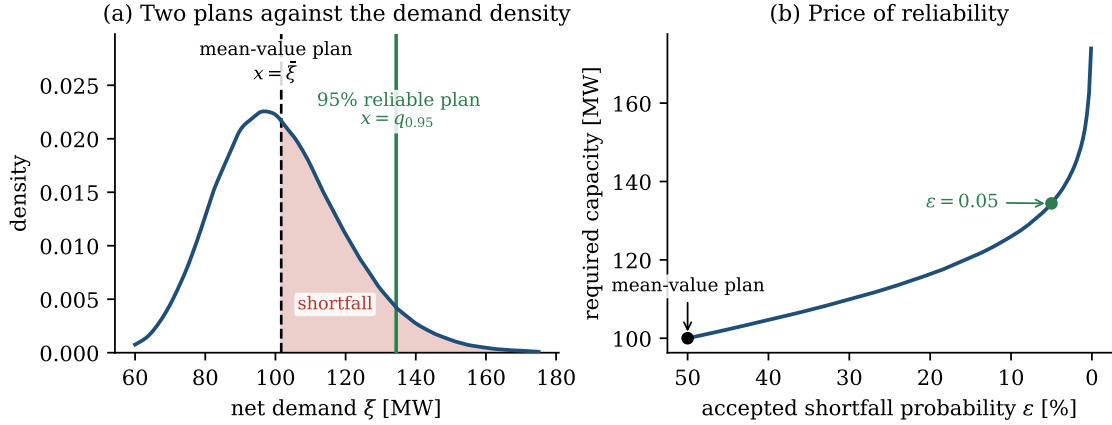


Figure 1: The mean-value fallacy in a one-dimensional dispatch problem with lognormal net demand (Monte Carlo with 2×10^5 samples). (a) The mean-value plan $x = \bar{\xi} \approx 102$ MW leaves the shaded shortfall region uncovered, so the constraint $\xi \leq x$ fails on 46% of days; covering demand with 95% reliability requires $x = q_{0.95} \approx 134$ MW. (b) Required capacity as a function of the accepted shortfall probability ε . Reliability is bought at an accelerating price as $\varepsilon \rightarrow 0$; the chance constraint makes this price explicit and lets the planner choose a point on the curve deliberately.

average capacity. Figure 1(a) shows what this decision looks like against the actual demand density. The constraint is violated on every day that demand exceeds its mean, and for this distribution that is 46% of all days. A plan that appears exact on paper fails almost every other day in operation.

The repair is equally easy to read off the figure. If the planner is willing to accept a shortfall on at most 5% of days, the appropriate decision is the 95% quantile of demand, $x = q_{0.95} \approx 134$ MW, about 32% more capacity than the mean-value plan. Figure 1(b) generalizes this observation into the central trade-off of the thesis: required capacity, and hence cost, rises steeply as the accepted violation probability ε shrinks, and rises without bound as $\varepsilon \rightarrow 0$ because the lognormal has unbounded support. Reliability is a commodity with a price, and the role of a properly formulated stochastic optimization problem is to make that price visible and chooseable rather than implicit and accidental.

1.1.4 A failure rate that recurs across problems

One might hope that the 46% figure is an artifact of the simple model. It is not. Figure 2 repeats the experiment for stylized versions of three of the domains above: dispatch under skewed demand, a landing-corridor constraint under bimodal thrust

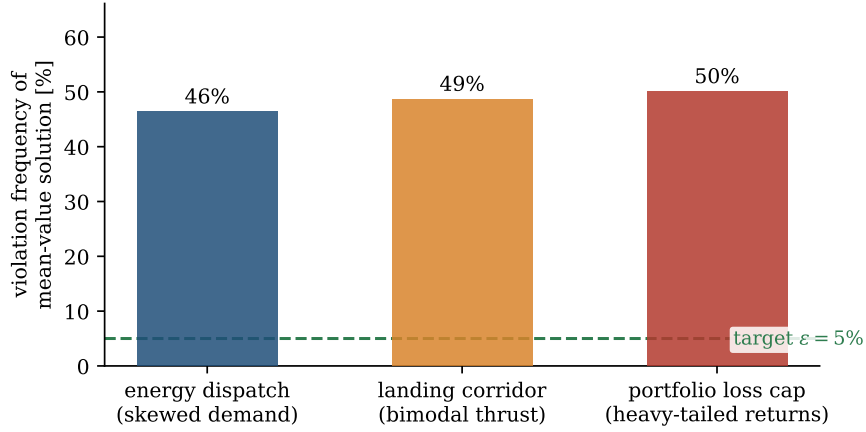


Figure 2: Empirical violation frequency of the mean-value solution in three stylized problems drawn from the application domains of this thesis (Monte Carlo estimates). A typical engineering target of $\varepsilon = 5\%$ is shown for scale. The recurring value close to 50% reflects the boundary mechanism: optimizers drive binding constraints to the edge of feasibility, where roughly half of all perturbations cause failure.

disturbance, and a portfolio loss cap under heavy-tailed (Student- t) returns. In every case the mean-value solution violates its constraint between 46% and 50% of the time, well beyond a typical design target. The violation frequency stays near one half largely because of the boundary mechanism described above: where the constraint is binding at the deterministic optimum, the median of the residual sits at or near zero, and this holds across the applications considered. The value therefore reflects a property of the optimization itself rather than of any particular application domain.

1.2 Two Extremes and the Gap Between Them

This section examines the two classical responses to the failure just described, the deterministic status quo and worst-case robustness, identifies the settings in which each is appropriate, and introduces the chance constraint as a formulation that connects the two through a tunable reliability level.

1.2.1 Ignoring uncertainty and worst-case robustness

Once the limitations of the mean-value approach are recognized, the question becomes what should replace it. Two answers dominate, and they sit at opposite ends of a spectrum.

The first answer is to ignore uncertainty entirely, which is the mean-value approach al-

ready examined in Section 1.1. Its defining property is worth restating in the language used for the rest of the thesis: the deterministic solution carries essentially no reliability statement. The formulation Eq. (1) does not bound the probability of failure, and Fig. 2 shows that this probability is typically near one half. Whatever reliability a deployed mean-value solution exhibits usually comes from safety margins added by hand, outside the optimization. Such hand-tuned margins can be read as an indication that the optimization, as posed, did not capture the intended requirement.

The second answer is worst-case robust optimization (Soyster, 1973; Ben-Tal u. a., 2009). The uncertain parameter is confined to an uncertainty set Ξ , and the constraint is enforced for every element of that set:

$$g(x, \xi) \leq 0 \quad \text{for all } \xi \in \Xi. \quad (3)$$

Robust optimization is a mature and elegant theory; for many set geometries, Eq. (3) admits exact tractable reformulations, and when a failure is not acceptable under any circumstance it is arguably the model of choice. Its weakness is the obverse of its strength: protection is purchased against every point of Ξ , including points that are vanishingly unlikely or physically implausible, and the objective pays for all of them. In the dispatch example, the demand distribution has unbounded support, so a literal worst-case formulation has no feasible solution at all; the set Ξ must be truncated by fiat, and the answer then depends sensitively on a truncation the data does not specify. Even with a defensible truncation, the protection level is largely a fixed property of the set rather than a dial the decision maker controls. Empirically, robust solutions in power systems and aerospace routinely sacrifice a large fraction of attainable objective value to guard against scenarios whose combined probability is far below any operational concern (Ben-Tal u. a., 2009; Keil u. a., 2021).

1.2.2 When each extreme is the right tool

Fairness demands an acknowledgment before the middle ground is claimed: each extreme has a legitimate home. The mean-value approach is correct when uncertainty is genuinely negligible relative to the constraint margins, or when violations carry only a linear cost that the objective already prices in, as in inventory models where a short-fall is simply an expense. Worst-case robustness is correct when the uncertainty set is small, well characterized, and physically bounded, and when a single failure would be severe and difficult to reverse: structural safety factors are a familiar example. The

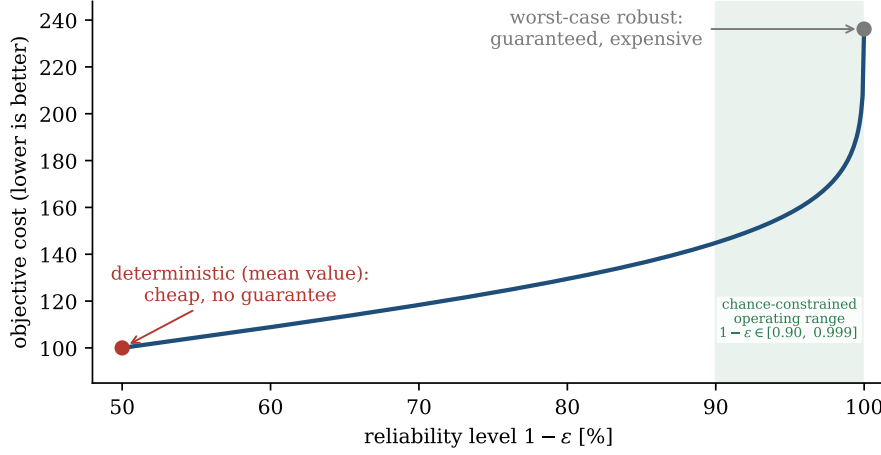


Figure 3: The reliability spectrum for the dispatch example of Fig. 1. The deterministic mean-value plan is cheap and offers no guarantee; the worst-case robust plan is guaranteed and expensive, and is not even well defined under unbounded support. Chance constraints parameterize the entire curve between the extremes: the decision maker selects a reliability level $1 - \varepsilon$, and the optimization returns the cheapest decision that honors it.

problems this thesis targets sit in neither home. Their uncertainties are large, data-described, and unbounded or effectively so, and their failures are costly but tolerable at a small, explicit rate. A lander program accepts a quantified risk per descent; a grid code tolerates a bounded loss-of-load expectation per year. For this regime, neither extreme expresses what the decision maker actually wants.

1.2.3 Chance constraints as the unifying middle ground

Between a method with no guarantee and a method that guarantees too much lies the formulation this thesis is about. A chance constraint, introduced by Charnes u. Cooper (1959) and developed into a full branch of stochastic programming by Prékopa (1995), requires the constraint to hold with a prescribed probability:

$$\mathbb{P}(g(x, \xi) \leq 0) \geq 1 - \varepsilon. \quad (4)$$

The risk tolerance $\varepsilon \in (0, 1)$ is small, typically between 0.001 and 0.1 depending on the stakes, and $1 - \varepsilon$ is the reliability level. Parts of the literature, including the convergence theory of Schuster (2025), write the required satisfaction probability directly as $\alpha = 1 - \varepsilon$; the two conventions are interchangeable, and this thesis works with ε throughout. Figure 3 places the three formulations on a single axis for the dispatch example. Rather than a compromise between the extremes, the chance constraint can

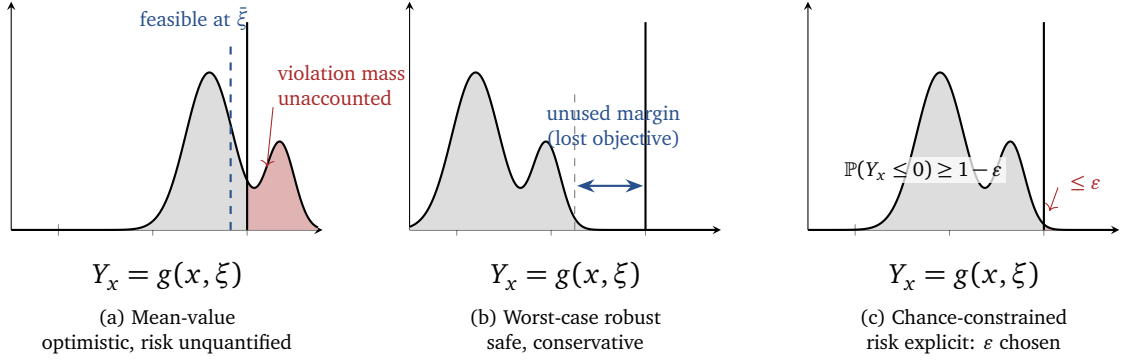


Figure 4: Three responses to constraint uncertainty, shown on the density of the residual $Y_x = g(x, \xi)$; the constraint is satisfied for $Y_x \leq 0$. **(a)** The mean-value solution satisfies the constraint at $\bar{\xi}$ but leaves an unquantified violation mass (red). **(b)** The worst-case robust solution eliminates all violations by pushing the distribution well clear of zero, at the cost of large unused margin. **(c)** The chance-constrained solution places exactly the permitted fraction ε of probability mass beyond the boundary, making the accepted risk explicit and controllable.

be viewed as generalizing both of them. As $\varepsilon \rightarrow 0$ it approaches worst-case protection over the support of ξ , and at $\varepsilon = 0.5$ it roughly reproduces the mean-value behavior for symmetric residuals. Every intermediate point is available, and each carries an explicit, auditable meaning: the system fails at most a fraction ε of the time. This closely mirrors the language in which engineering requirements are commonly written. Grid codes prescribe loss-of-load expectations, flight rules prescribe acceptable failure probabilities per descent, and risk mandates prescribe loss probabilities per quarter. The chance constraint is the optimization-native translation of a reliability requirement, which is why it is a natural framing for many engineering and operational problems, and why the difficulty of solving it, the subject of Section 1.3, has remained an active research frontier for over six decades. Figure 4 makes the three positions concrete. Each panel shows the same bimodal residual density $Y_x = g(x, \xi)$, with the constraint boundary at zero; the two modes reflect the non-Gaussian disturbance shapes that motivate the nonparametric approach developed from subsection 1.4 onward

1.3 What Chance Constraints Are and Why They Are Hard

This section fixes the notation used throughout the thesis, formalizes individual and joint chance constraints, and isolates the three structural properties that make probabilistic constraints computationally demanding. These three difficulties also organize the survey of solution methods in Chapter 2.

1.3.1 Individual and joint chance constraints

This section fixes notation used throughout the thesis and isolates the three structural sources of computational difficulty. Let $x \in X \subseteq \mathbb{R}^n$ be the decision vector, let ξ be a random vector on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ taking values in $\mathbb{R}^d$, and let $g : \mathbb{R}^n \times \mathbb{R}^d \rightarrow \mathbb{R}$ be a constraint function. The *individual chance constraint* is

$$\mathbb{P}(g(x, \xi) \leq 0) \geq 1 - \varepsilon. \quad (5)$$

When the system carries several uncertain requirements $g_1, \dots, g_R$ that must hold simultaneously, the appropriate model is the *joint chance constraint*

$$\mathbb{P}(g_r(x, \xi) \leq 0, r = 1, \dots, R) \geq 1 - \varepsilon, \quad (6)$$

which bounds the probability that *all* requirements hold at once. The distinction matters: enforcing each g_r individually at level $1 - \varepsilon$ does not bound the joint failure probability, while the classical repair via Boole's inequality, splitting the budget as $\sum_r \varepsilon_r \leq \varepsilon$, is sound but conservative (Prékopa, 1995; Keil u. a., 2021). The full chance-constrained program studied in this thesis is

$$\underset{x \in X}{\text{minimize}} \ f(x) \quad \text{subject to} \quad \mathbb{P}(g(x, \xi) \leq 0) \geq 1 - \varepsilon. \quad (7)$$

A word on the role of ε is warranted, because it is the one quantity in Eq. (7) that mathematics cannot supply. The risk tolerance is a design parameter set by the consequences of failure and by external requirements: a 5% daily shortfall tolerance may be acceptable for a microgrid with backup supply (Ciftci u. a., 2019), while a descent constraint may demand $\varepsilon \leq 10^{-2}$ or far less (Blackmore u. a., 2010). Two facts about this parameter shape everything downstream. First, the difficulty of the problem grows as ε shrinks, because the constraint then probes ever deeper into the tail of a distribution that must be known or estimated ever more accurately there. Second, ε is a budget, and for joint constraints Eq. (6) the budget must cover all R requirements simultaneously, which is what makes the joint form strictly harder than the individual one. Throughout the thesis, ε is treated as given and the question is how to honor it efficiently and honestly.

1.3.2 Three structural sources of difficulty

Writing Eq. (7) is easy; solving it is not, and the obstruction is threefold. Each difficulty below is independent of the others, and any viable solution method must answer all three. The pattern repeats through the entire methodological literature, so these three points are the lens through which every method in this thesis, including the KDE reformulation itself, should be judged.

Difficulty 1: the feasible set is generally non-convex. Convexity of g in x does not transfer to the probabilistic feasible set $\{x : \mathbb{P}(g(x, \xi) \leq 0) \geq 1 - \varepsilon\}$. A minimal example makes the geometry vivid. Take a linear constraint $\xi^\top x \leq 1$ where ξ takes one of two values $\xi^{(1)}, \xi^{(2)}$ with probability one half each, and set $\varepsilon = 0.5$. The chance constraint then requires $\xi^\top x \leq 1$ for *at least one* of the two scenarios, so the feasible set is the union of two half-spaces. Figure 5(a) shows the result: two points, each satisfying a different scenario, are individually feasible, while their midpoint satisfies neither and is infeasible. A union of convex sets is not convex, and this toy construction scales up: with general discrete or continuous distributions the feasible set can shatter into combinatorially many pieces. Convexity survives only under special structure, most famously when the distribution of ξ is log-concave and g has suitable form, the well-known theory of Prékopa (1995), or when ξ is Gaussian and the constraint is linear in ξ , in which case Eq. (5) collapses to a second-order cone constraint (Charnes u. Cooper, 1963; Shapiro u. a., 2014). Outside these islands, non-convexity is the rule, and global optimality claims must be abandoned in favor of high-quality local solutions, which is precisely the regime in which nonlinear programming solvers operate.

Difficulty 2: the probability has no closed form. Define the parameterized failure region $M(x) = \{\xi : g(x, \xi) > 0\}$. Evaluating the constraint in Eq. (7) means computing

$$\mathbb{P}(g(x, \xi) \leq 0) = 1 - \int_{M(x)} \rho_\xi(\xi) d\xi, \quad (8)$$

an integral of the density ρ_ξ over a region whose shape changes with every candidate decision x . For nonlinear g the region $M(x)$ is curved and possibly disconnected, the integral is multidimensional, and a closed form exists only in textbook cases. Worse, in the data-driven settings that motivate this thesis the density ρ_ξ itself is unknown; all that exists is a finite collection of samples from simulation, historical records, or exper-

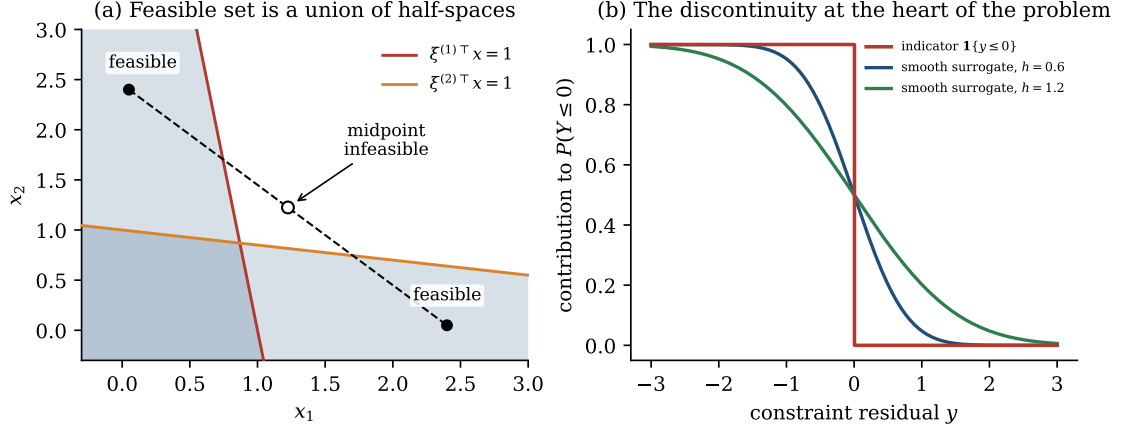


Figure 5: The two geometric faces of chance-constraint hardness. (a) For a two-point distribution of ξ and $\varepsilon = 0.5$, the feasible set of $\mathbb{P}(\xi^\top x \leq 1) \geq 0.5$ is the union of two half-spaces (shaded). The marked endpoints are feasible while their midpoint is not, certifying non-convexity. (b) The contribution of a single sample to the probability is the discontinuous indicator $\mathbf{1}\{y \leq 0\}$ of the residual; its derivative is zero almost everywhere and undefined at the jump. Smooth surrogates obtained by integrating a kernel, shown for two bandwidths h , restore the differentiability that gradient-based solvers require, at the price of a bias controlled by h . This single picture contains the central tension of the thesis.

iments. The constraint function that the optimizer must evaluate, thousands of times, at thousands of points x , is therefore an unknown integral of an unknown density over an intractable region. Any practical method replaces Eq. (8) with an approximation, and the quality of that approximation governs everything downstream.

Difficulty 3: the probability is built from a discontinuous function. Rewriting the probability as an expectation,

$$\mathbb{P}(g(x, \xi) \leq 0) = \mathbb{E}[\mathbf{1}\{g(x, \xi) \leq 0\}], \quad (9)$$

exposes the third obstruction: the indicator $\mathbf{1}\{\cdot\}$ jumps from one to zero at the constraint boundary, as plotted in Fig. 5(b). The natural sample-based estimate of Eq. (9), count the fraction of samples satisfying the constraint, inherits this discontinuity. As a function of x it is piecewise constant: its gradient is zero almost everywhere and undefined on a measure-zero set, which is poorly suited to the sequential quadratic programming and interior-point methods that are widely used in nonlinear optimization (Shapiro u. a., 2014). The established escape routes each pay a recognizable price. Mixed-integer reformulations of the counting constraint are exact but combinatorial and scale poorly (Luedtke u. Ahmed, 2008). The scenario approach enforces the

constraint on all N sampled scenarios, regaining convexity and even finite-sample feasibility certificates for convex problems (Calafiore u. Campi, 2006; Campi u. Garatti, 2008), but it cannot exploit the slack that ε permits and grows conservative as N increases. Conservative convex surrogates such as conditional value-at-risk restrict the feasible set from inside (Rockafellar u. Uryasev, 2000; Nemirovski u. Shapiro, 2006), guaranteeing feasibility at a cost in objective value. Smoothing methods, the family to which this thesis belongs, replace the indicator with a differentiable approximation (Peña-Ordieres u. a., 2020), restoring the gradients that nonlinear solvers need while introducing a controlled bias, visible in Fig. 5(b) as the gap between the step and the smooth curves.

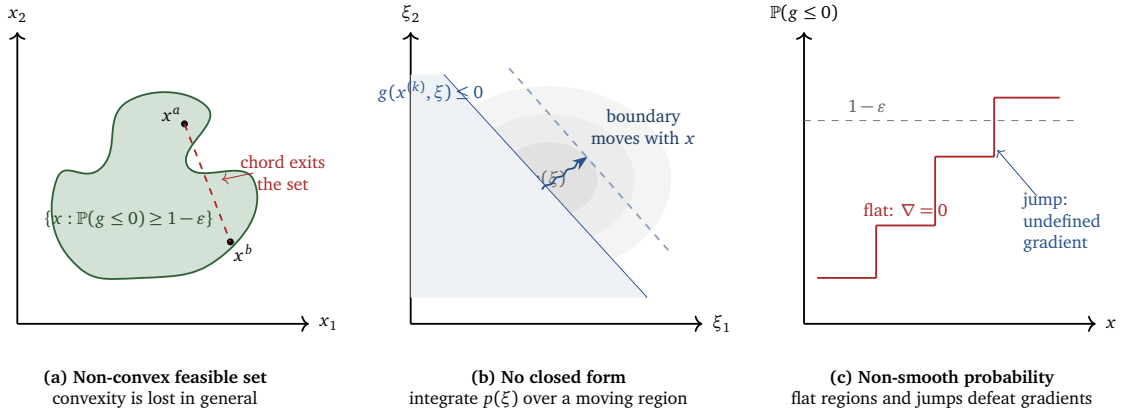


Figure 6: The three structural sources of difficulty in chance-constrained programming: (a) the feasible set is non-convex, (b) the constraint probability has no closed form because the safe region moves with x , and (c) the probability function is flat almost everywhere and non-differentiable at the boundary.

The combination of the three difficulties, and not any one of them alone, is what makes Eq. (7) substantially harder than a deterministic NLP. Non-convexity generally precludes global guarantees, the missing closed form precludes exact evaluation, and non-smoothness makes naive sample averaging unsuitable inside derivative-based solvers. Every approximation method discussed in this thesis is an attempt to buy back tractability by conceding something: exactness, generality, or a margin of conservatism. The contribution of the KDE-based reformulation is a specific and unusually favorable bargain on all three fronts, and the remainder of this chapter explains why.

1.4 The Case for Nonparametric Estimation

This section develops the argument for estimating distributions from data. It reviews the parametric alternative and the evidence against its default assumptions, quantifies the consequences of a misspecified distribution, introduces kernel density estimation, and closes with an account of the costs that the nonparametric route incurs.

1.4.1 The parametric assumption and its routine failure

Among the classical escape routes from Section 1.3, the oldest and still most widely used is parametric: assume the law of ξ belongs to a named family, almost always the Gaussian, and exploit the structure that follows. Under Gaussian ξ and a constraint linear in ξ , the chance constraint Eq. (5) reduces exactly to a deterministic second-order cone constraint involving the mean, the covariance, and a quantile of the standard normal (Charnes u. Cooper, 1963; Shapiro u. a., 2014). The reformulation is convex, smooth, and supported by every modern conic solver. When the Gaussian assumption holds, this reformulation is difficult to improve upon, and the experimental chapters use it as the reference baseline for that reason.

The assumption, however, is routinely false in the very domains where chance constraints matter most, and the evidence is not anecdotal. Wind-power forecast errors are skewed and heavy-tailed, with documented deviations from normality that grow in the tails, which is precisely where a chance constraint lives (Hodge u. Milligan, 2011; Wan u. a., 2020). Equity and commodity returns exhibit heavy tails and aggregational Gaussianity only at long horizons; at decision-relevant horizons their kurtosis is far above three (Cont, 2001). Thrust disturbances and navigation errors in powered descent arise from mixtures of operating regimes and are naturally multimodal (Keil u. a., 2021). Process feedstocks vary by supplier and batch in ways no single unimodal family captures (Calfa u. a., 2015). The Gaussian is often chosen more for analytical convenience than for its fit to the observed behavior of the system.

1.4.2 What a wrong distribution does to the guarantee

The consequences of the mismatch are asymmetric and quantifiable. Figure 7 dissects both failure modes. Panel (a) shows a bimodal disturbance law next to its moment-matched Gaussian fit: the fit invents probability where there is none and erases the two modes that actually govern system behavior. Panel (b) shows the more dangerous

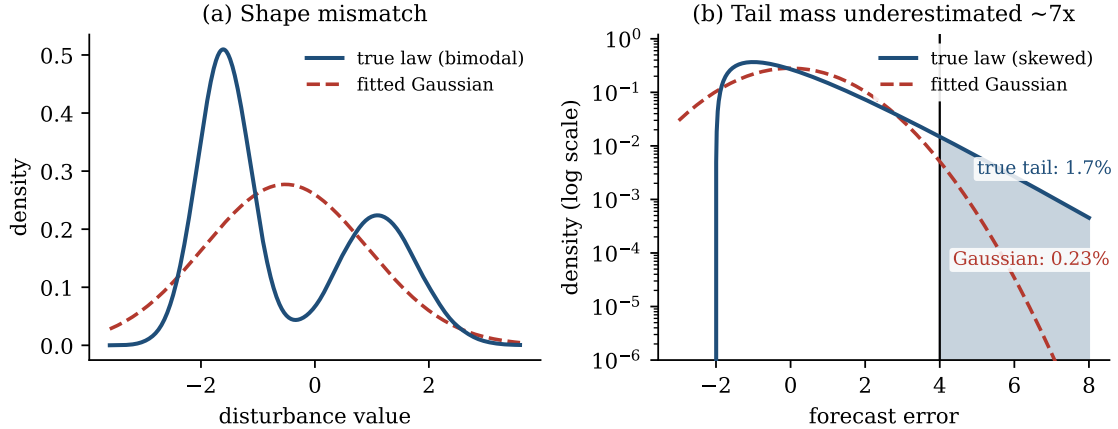


Figure 7: What a wrong distributional assumption does to a chance constraint. (a) A bimodal disturbance law, of the kind produced by two-regime actuator behavior, against its best Gaussian fit: the fit places probability mass between the modes where the true law has almost none, and misplaces both shoulders. (b) A skewed forecast-error law (centered Gamma) on a logarithmic density scale. Beyond the marked threshold the Gaussian fit reports 0.23% tail mass where the true law carries 1.7%, an underestimate by a factor of about seven. A planner who certifies a 0.5% risk level with the Gaussian model deploys a system whose true failure rate is more than three times the certificate.

case on a logarithmic scale. For a skewed forecast-error law, the Gaussian fit underestimates the mass beyond a critical threshold by a factor of seven: it reports 0.23% where the truth is 1.7%. A decision certified to a 0.5% failure probability under the Gaussian model in fact fails more than three times as often as certified. The error can also point the other way: when the true law is lighter-tailed than the fit, the parametric constraint is needlessly conservative and pays an invisible objective penalty. Whether the result is optimistic or conservative, the direction is governed by distributional geometry that the parametric fit does not capture, and the certificate that the chance constraint was intended to provide, $\mathbb{P}(\text{failure}) \leq \varepsilon$, may no longer hold. A reliability guarantee that rests on an unverified distributional assumption therefore provides considerably weaker assurance than its form suggests.

1.4.3 Kernel density estimation and its fit to optimization

The alternative pursued in this thesis is to estimate the distribution from the data instead of assuming it. Kernel density estimation, the canonical nonparametric estimator introduced by Rosenblatt (1956) and Parzen (1962), reconstructs a density

from samples $y_1, \dots, y_N$ as

$$\hat{\rho}_h(y) = \frac{1}{Nh} \sum_{j=1}^N K\left(\frac{y - y_j}{h}\right), \quad (10)$$

where the kernel K is a smooth density and the bandwidth $h > 0$ sets the degree of smoothing. The estimator makes no shape commitment: given enough data it recovers skewness, multimodality, and heavy tails alike, with well-understood consistency theory and practical bandwidth rules (Silverman, 1986; Devroye u. Györfi, 1985). Three properties make KDE specifically suited to the optimization context, rather than merely to statistics. First, $\hat{\rho}_h$ is as smooth as its kernel, so quantities built from it are differentiable, which addresses Difficulty 3 head-on. Second, its integral is available in closed form whenever the kernel's cumulative distribution function is, which addresses the evaluation half of Difficulty 2. Third, it is distribution-free, which removes the implicit distributional assumption discussed above.

This is not a speculative combination. A coherent body of work already demonstrates KDE-based chance constraints across the application domains of Section 1.1: kernel-smoothed individual and joint chance constraints in process planning (Calfa u. a., 2015), KDE-reformulated optimal control for launcher ascent and the Goddard problem (Caillau u. a., 2018), biased-KDE optimal control for soft lunar landing (Keil u. a., 2020, 2021; Keil u. Rao, 2021), adaptive-KDE microgrid energy management (Ciftci, 2017; Ciftci u. a., 2019; Babić u. a., 2023), multivariate-KDE joint chance constraints for economic dispatch and power-system scheduling (Wu u. a., 2021; Wu u. Kargarian, 2023), KDE-driven distributionally robust load shedding (Qi u. a., 2026), and KDE-based tail-risk portfolio optimization (Liu u. a., 2022). Adjacent nonparametric formulations reinforce the same direction in robotics and forecasting (Gopalakrishnan u. a., 2018; Wan u. a., 2020; Wang u. a., 2021). What this literature has lacked, until recently, is a unifying account connecting the engineering constructions to convergence theory; supplying and stress-testing that connection is the work of this thesis.

1.4.4 The price of going nonparametric

A nonparametric estimator does not come free, and the costs are named now because they structure the experimental program of the application chapter. KDE suffers the curse of dimensionality: its convergence rate degrades rapidly as the dimension of the estimated variable grows (Silverman, 1986). The bandwidth h is a genuine hyperpa-

parameter whose role inside an optimization problem, where it shapes the feasible region and the gradients the solver sees, is more delicate than its role in density estimation. A plain KDE smooths in both directions, so it can overestimate the safe probability and thereby weaken the very bound it is meant to enforce. A fourth cost concerns the tails: a chance constraint with a small violation budget asks about a rare event, and a kernel estimate is least accurate in sparsely sampled tail regions. The severity of this effect is problem dependent rather than universal, since at moderate risk levels the binding region lies in the bulk of the distribution, where the estimator behaves well, while stringent levels push it into the tails. Each of these costs has a specific answer in this thesis: the first is mitigated by the residual-space construction of Section 1.5, the second is treated as a first-class experimental question, the third is addressed by the biased-KDE construction of Keil u. a. (2021), and the fourth is handled by reporting results across a range of risk levels.

1.5 The Residual-Space Insight

This section introduces the change of variables on which the methodological part of the thesis is built. It defines the constraint residual, explains why estimating its distribution is preferable to estimating the distribution of the uncertainty itself, and previews the treatment of joint constraints together with the role of the bandwidth.

1.5.1 From uncertainty space to residual space

Everything assembled so far points toward a single conceptual move, simple to state and central to the approach developed in this thesis. The naive way to use KDE in Eq. (7) is to estimate the density of the uncertainty ξ itself and then integrate the estimate over the safe region $\mathbb{R}^d \setminus M(x)$. This inherits both halves of Difficulty 2: the estimation runs in d dimensions, exactly where the curse of dimensionality bites, and the integration region still deforms with every candidate x . A more effective object to estimate is not the uncertainty itself but the *constraint residual*

$$Y(x) = g(x, \xi), \tag{11}$$

the scalar random variable obtained by pushing the uncertainty through the constraint function at the current decision. The residual is the quantity the constraint actually speaks about: $Y(x) \leq 0$ is constraint satisfaction, by definition, and the chance con-

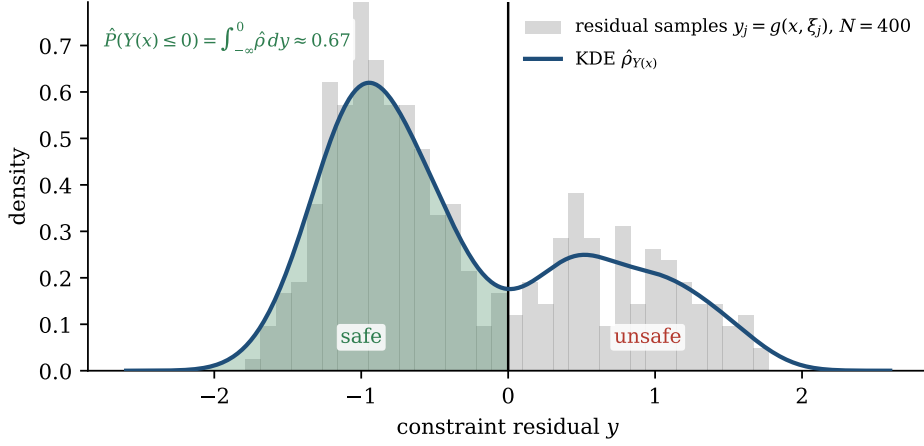


Figure 8: The residual-space construction at a fixed candidate decision x . Samples ξ_j of the uncertainty are pushed through the constraint to give residual samples $y_j = g(x, \xi_j)$ (histogram, $N = 400$, bimodal by construction); a KDE $\hat{\rho}_{Y(x)}$ smooths them, and its integral over the fixed safe region $(-\infty, 0]$ (shaded) estimates the safe probability, here $\hat{P} \approx 0.67$. The estimation problem is one-dimensional regardless of the dimension of ξ , and the resulting constraint $\hat{F}_{Y(x)}(0) \geq 1 - \varepsilon$ is a smooth, differentiable function of x .

straint Eq. (5) is nothing but a statement about the cumulative distribution function of $Y(x)$ at zero,

$$\mathbb{P}(Y(x) \leq 0) = F_{Y(x)}(0) \geq 1 - \varepsilon. \quad (12)$$

Schuster (2025) makes this change of variables the foundation of the convergence theory: the probability is rewritten as an integral of the density of the *transformed* variable $g(x, \xi)$ over the fixed set $(-\infty, 0]$, rather than an integral of the density of ξ over the moving set $\mathbb{R}^d \setminus M(x)$.

1.5.2 Three dividends of the change of variables

The benefits of this shift are worth enumerating, because each one addresses a specific difficulty from Section 1.3.

First, *dimensionality collapses*. The uncertainty ξ may live in ten or a hundred dimensions; the residual of an individual constraint is a scalar, always, and the residuals of a joint constraint form a vector of dimension R , the number of constraint components, which in the problems of this thesis is small. This is where the *curse of dimensionality*, named as the first cost of nonparametric estimation in Section 1.4.4, is addressed. The convergence rate of a kernel estimator deteriorates rapidly with the dimension of the estimated variable, so that holding the accuracy fixed requires a sample size

that grows roughly exponentially in that dimension (Silverman, 1986). The decisive observation is that the relevant dimension for the residual-space construction is the dimension of the estimated variable, not of the uncertainty: KDE is asked to perform one-dimensional density estimation from a few hundred samples, the task it is best suited to, and the curse in d is largely avoided by construction. The constraint function g itself performs the dimension reduction, and it does so without information loss for the question being asked, since Eq. (12) is an identity, not an approximation. The mitigation is not unconditional, however; it weakens when many constraints must hold jointly, since the residual vector then carries the dimension R , a point taken up in Section 1.5.3.

Second, *the integration region freezes*. The safe region in residual space is $(-\infty, 0]$ for every decision x ; all dependence on x has migrated into the samples $y_j = g(x, \xi_j)$ themselves. Integrating the kernel estimate over a fixed half-line is elementary: substituting Eq. (10) gives the estimated CDF

$$\hat{F}_{Y(x)}(0) = \frac{1}{N} \sum_{j=1}^N \mathcal{K}\left(\frac{-g(x, \xi_j)}{h}\right), \quad \mathcal{K}(t) = \int_{-\infty}^t K(s) ds, \quad (13)$$

a finite sum of evaluations of the kernel's CDF $\mathcal{K}$, available in closed form for every kernel used in this thesis.

Third, *smoothness is restored*. Comparing Eq. (13) with the sample-average form Eq. (9) reveals the substitution at the heart of the method: the discontinuous indicator $\mathbf{1}\{g(x, \xi_j) \leq 0\}$ has been replaced by the smooth sigmoid $\mathcal{K}(-g(x, \xi_j)/h)$, exactly the curves drawn in Fig. 5(b). If g is differentiable in x , then so is $\hat{F}_{Y(x)}(0)$, with gradients obtained by the chain rule through $\mathcal{K}$. The chance constraint Eq. (5) thereby becomes the deterministic, smooth, nonlinear constraint

$$\hat{F}_{Y(x)}(0) \geq 1 - \varepsilon, \quad (14)$$

which any sequential quadratic programming, interior-point, or direct-collocation solver consumes natively. Figure 8 shows the whole construction in one frame: residual samples, the kernel estimate, the frozen safe region, and the resulting probability estimate.

1.5.3 Joint constraints, bandwidth, and the safety guarantee

The construction extends to joint chance constraints with one further idea that previews the extension studied in this thesis. For Eq. (6), the natural scalar summary of the constraint system is the worst violation, $Y(x) = \max_r g_r(x, \xi)$, since all requirements hold exactly when the maximum residual is nonpositive. The maximum preserves the one-dimensionality of the estimation problem but reintroduces a kink; smoothing it with a log-sum-exp surrogate restores differentiability at the cost of a second, controllable approximation layer. This scalar violation statistic, its interaction with the KDE layer, and the alternative of low-dimensional multivariate KDE over the residual vector form the substance of research question RQ4 and are developed in Part 3.

The reformulation Eq. (14) is where the thesis begins rather than ends, because the two costs flagged in Section 1.4 are still standing and they define the methodological arc of Part 3. The bandwidth h now controls the geometry of an optimization problem, not just the fidelity of a density plot: $h \rightarrow 0$ recovers the non-smooth indicator and its numerical pathologies, while large h oversmooths and distorts the feasible region, and no classical bandwidth rule was derived with either effect in mind. And because an ordinary KDE can overestimate $F_{Y(x)}(0)$, the constraint Eq. (14) can certify a reliability the true system does not possess. The constructive answer is the biased KDE of Keil u. a. (2020, 2021): kernels engineered so that their integral function bounds the indicator from the safe side, making Eq. (14) a guaranteed conservative approximation of the true chance constraint while remaining smooth and bounded. The theoretical answer is the convergence framework of Schuster (2025), which establishes conditions under which solutions of the KDE-approximated problem converge to solutions of the true problem as $N \rightarrow \infty$. Engineering construction and convergence theory, developed independently, meet in residual space, and their meeting is the subject of this thesis.

1.6 Thesis Scope and Positioning

This closing section delimits the scope of the thesis. It positions the work between its two anchor papers, states which questions lie outside the scope, and formulates the four research questions together with a roadmap of the remaining chapters.

1.6.1 Disciplinary positioning and anchor literature

A thesis at the intersection of three fields must say plainly what it is and what it is not. This thesis is not a contribution to nonparametric statistics: it proves no new properties of kernel density estimators, and it takes the consistency theory of Parzen (1962); Silverman (1986); Devroye u. Györfi (1985) as given. It is not a contribution to control theory: it establishes no stability or recursive feasibility results for closed-loop predictive control. It is a data science and applied optimization contribution. Its object of study is the KDE-based reformulation pipeline itself, from samples through residuals through Eq. (14) into a numerical solver, and its method is to implement that pipeline rigorously, analyze its behavior, compare it against the established alternatives, and validate it across problem settings chosen to stress its assumptions. One further boundary deserves explicit mention. Every guarantee developed in this thesis is a statement about the distribution from which the samples were drawn. Whether such a guarantee transfers to the distribution encountered in deployment, when samples come from a simulator or from historical records that represent the target system imperfectly, is a separate question that this thesis does not address; distributionally robust formulations (Chen u. a., 2024) are a natural direction for that question.

Two anchor papers bound the territory. Keil u. a. (2021), with the earlier preprint Keil u. a. (2020) and the computational follow-up Keil u. Rao (2021), supply the constructive, engineering side: biased kernels whose integral functions bound the indicator, a safety-preserving deterministic reformulation, and a demonstration on chance-constrained soft lunar landing solved by direct collocation. Schuster (2025) supplies the analytical side: convergence theorems guaranteeing that, under verifiable conditions on the kernel, the bandwidth, and the problem, solving the estimated problem is asymptotically equivalent to solving the true one. The first anchor gives the method its practical credibility, the second its mathematical license; the thesis builds the bridge between them and load-tests it experimentally. The surrounding evidence base spans aerospace (Caillaud u. a., 2018), energy systems (Ciftci u. a., 2019; Wu u. a., 2021; Wu u. Kargarian, 2023; Qi u. a., 2026), process systems (Calfa u. a., 2015), and finance (Liu u. a., 2022), and the comparative baselines are the classical ones: scenario optimization (Calafiore u. Campi, 2006; Campi u. Garatti, 2008), sample average approximation (Luedtke u. Ahmed, 2008; Pagnoncelli u. a., 2009), parametric reformulation (Charnes u. Cooper, 1963), and convex risk surrogates (Rockafellar u. Uryasev, 2000; Nemirovski u. Shapiro, 2006). Further smooth alternatives exist for structured settings, notably polynomial chaos expansions (Xiu u. Karniadakis, 2002)

and spheric-radial gradient formulae for Gaussian-like laws (van Ackooij u. Henrion, 2014); a systematic benchmark against these lies outside the scope of this thesis and is noted as future work.

1.6.2 Research questions and roadmap

Four research questions organize everything that follows.

- **RQ1: formulation** how is a chance-constrained problem reformulated through KDE over constraint residuals, for individual and joint constraints, and under what conditions is the reformulation mathematically justified?
- **RQ2: numerical behavior** how does the KDE reformulation compare with deterministic, scenario-based, and parametric approaches in feasibility, objective value, conservatism, runtime, and sensitivity to sample size?
- **RQ3: bandwidth and smoothing** how do bandwidth and smoothing choices shape feasibility, conservatism, and solver performance, given that h acts on the optimization geometry and not only on estimation error?
- **RQ4: extension** can the framework be strengthened through one focused extension, specifically scalar violation statistics for joint chance constraints, optionally with log-sum-exp smoothing?

The argument now proceeds in three movements. Part 2 supplies the technical vocabulary: the taxonomy of optimization under uncertainty, the theory of kernel density estimation, and the elements of numerical optimal control. Part 3 develops the mathematics in the order a practitioner would discover it: the blackbox sample-counting approach and its failure inside gradient-based solvers, the smooth unbiased KDE reformulation, the biased construction that restores the safety guarantee, and the convergence theory that certifies the whole. Part 4 validates the framework experimentally, from controlled benchmarks through the lunar landing problem, and distills the results into practical guidance. The aim is a defensible answer to a question that each of the systems in Section 1.1 raises implicitly: how safe is safe enough, and what is the cheapest decision that reliably delivers it?

2 Stochastic Optimization and Chance Constraints

Chapter 1 settled two points this chapter builds on. First, the chance constraint is the right way to express reliability requirements, since it is the only formulation in which the decision maker sets the protection level directly. Second, it is hard to enforce, for three structural reasons: a usually non-convex feasible set, a probability with no closed form, and an indicator function with no usable derivative. Chapter 1 explained why this matters; this chapter supplies the tools.

The order is deliberate, as each part prepares the next. Section 2.1 sets out the taxonomy of optimization under uncertainty, since the special place of chance constraints only makes sense once the alternatives are defined. Section 2.2 presents the classical theory and the cases where convexity holds, showing how little of that structure survives under realistic distributions. Section 2.3 then surveys six approximation methods, each described through its core idea, main advantage, and main limitation, and each answering a weakness of the previous one.

With the gap identified, Section 2.4 introduces the tool used to fill it, kernel density estimation, presented from first principles because optimization readers will rarely have seen it this way. Section 2.5 closes by drawing the existing work that combines KDE with chance constraints into a single account, whose one open thread is this thesis.

2.1 A Taxonomy of Optimization Under Uncertainty

2.1.1 Four paradigms

Every approach to optimization under uncertainty answers two questions: how much information about the random parameter ξ the model uses, and how much feasibility against ξ the solution guarantees. These two axes are not just a way of organizing the discussion; they are how the field actually divides. Four answers cover the classical landscape, and fixing them precisely now is what turns the claim made for chance constraints in Section 2.1.2 into a real statement.

Deterministic optimization uses the least information and guarantees the least feasibility. The random parameter is replaced by a single representative value, usually a mean or nominal forecast, and the problem is solved as if that value were certain. Its formulation, failure mechanism, and near 50% violation rates were treated in Chap-

ter 1 and need no repetition; here it is simply the zero point against which the other paradigms are measured.

Robust optimization uses set information. The modeler specifies an uncertainty set Ξ assumed to contain every relevant realization, and feasibility is enforced for all of them at once, $g(x, \xi) \leq 0$ for all $\xi \in \Xi$ (Soyster, 1973; Ben-Tal u. a., 2009). The problems are minimax, and a mature duality theory turns many into tractable deterministic equivalents. Feasibility is maximal over Ξ but all-or-nothing. Every point is covered equally, including points of vanishing probability, and the objective pays for all of them.

Stochastic programming uses the full distribution and optimizes in expectation. In its two-stage form, a here-and-now decision x is taken before ξ is revealed and a recourse decision $y(\xi)$ repairs the outcome afterwards, with objective $f(x) + \mathbb{E}[Q(x, \xi)]$, where Q is the optimal recourse cost (Dantzig, 1955; Birge u. Louveaux, 2011). It is the natural choice when violations are costs rather than failures. Its coverage is in turn soft. The solution is good on average, but nothing guarantees the probability that a given realization behaves poorly.

Chance-constrained programming, the framework of this thesis, also uses the complete distribution (or samples from it), but spends that information on a hard probabilistic guarantee rather than an average: $\mathbb{P}(g(x, \xi) \leq 0) \geq 1 - \varepsilon$ (Charnes u. Cooper, 1959; Prékopa, 1995). In stochastic programming, violations are treated as costs. In chance constraints, violations are limited directly. The allowed limit, represented by ε , is chosen by the decision maker. This approach is appropriate when violations are clear failure events, such as a crash, load shedding, or a risk-limit breach, and the main concern is how often they occur.

2.1.2 A conceptual map and where this thesis sits

Figure 9 places the four paradigms on the two axes above. Reading the map gives three observations that recur throughout the thesis. First, the paradigms are not competitors on a single scale; they sit at different points of a two-dimensional trade-off, and each is the right tool somewhere, as Chapter 1 noted for the two extremes. Second, chance-constrained programming is the only paradigm whose protection level is a continuous dial rather than a fixed point: moving ε from 0.5 toward 0 shifts the solution continuously from mean-value behavior toward worst-case behavior over the support. Third, this dial comes at the price identified in Chapter 1: the probability operator destroys the convexity, evaluability, and smoothness that the other three

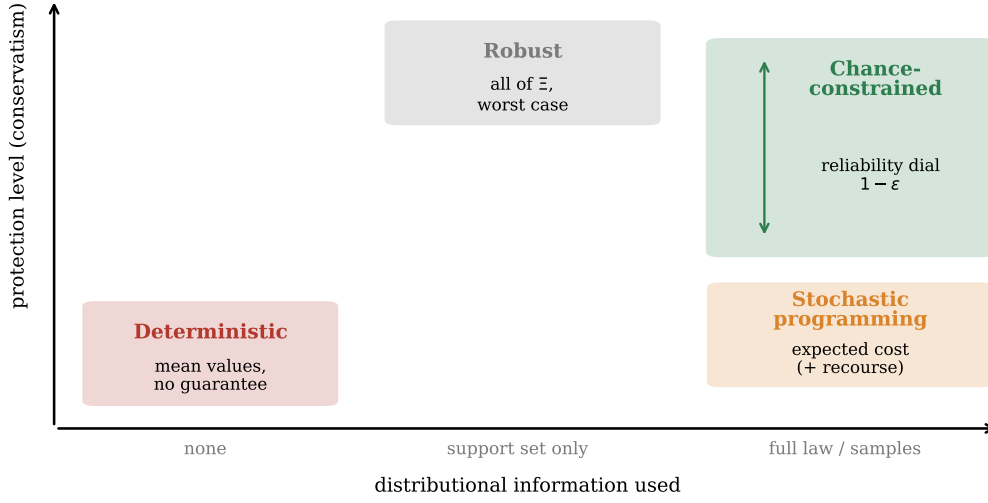


Figure 9: The four paradigms of optimization under uncertainty arranged by the distributional information they consume (horizontal) and the protection they provide (vertical). Deterministic optimization uses point estimates and guarantees nothing; robust optimization defends a whole set at maximal conservatism; stochastic programming optimizes expected cost, possibly with recourse, without bounding individual failure probabilities. Chance-constrained programming is the only paradigm with a continuously tunable protection level, the reliability dial $1 - \epsilon$.

paradigms keep, which is why the approximation methods of Section 2.3 are needed at all.

Two remarks complete the map. Hybrid formulations exist and are actively studied: distributionally robust chance constraints enforce the guarantee for every distribution in an ambiguity ball around the empirical law, combining the chance-constrained dial with robust-style protection against estimation error (Chen u. a., 2024; Liu u. a., 2022). They return in Section 2.3.7 as a row of the comparison table. Also, the boundaries between paradigms are modeling choices, not laws of nature: the same problem can often be written in several of them, and part of the modeler’s craft is matching the paradigm to the reliability language of the application. The applications of this thesis, surveyed in Chapter 1, all speak in failure probabilities, which fixes the choice.

2.2 Classical Chance-Constrained Programming

The taxonomy assigned chance-constrained programming its position; this subsection establishes its content. The order of business is the one any mature theory imposes: first the canonical formulations and the equivalent forms the rest of the thesis com-

putes with, then the strongest structural result the field possesses, and finally an honest accounting of how far that result reaches. The accounting matters, because the methods of Section 2.3 exist precisely in the territory the classical theory does not cover, and the size of that territory is the size of the problem this thesis addresses.

2.2.1 Origins and canonical formulations

Chance constraints were not invented as mathematics; they were forced into existence by a scheduling problem. Planning heating-oil production at a refinery, Charnes u. Cooper (1959) faced demands that no fixed plan could satisfy on every realization, and drew the conclusion the field has operated on ever since: a planning constraint need not hold always, only sufficiently often, and the required frequency belongs in the model as a first-class quantity. Charnes u. Cooper (1963) supplied the first deterministic equivalents for the resulting programs, and Miller u. Wagner (1965) extended the framework from individual to joint constraints, recognizing that a system of requirements demands a probability statement about the system, not about each row separately. The decisive consolidation came with Prékopa, whose log-concavity theory (Section 2.2.2) and monograph (Prékopa, 1995) elevated a modeling device into a branch of mathematical programming with its own structural theorems. The formulations fixed in Chapter 1 are restated here in the notation used for the remainder of the thesis. With decision $x \in X \subseteq \mathbb{R}^n$, random parameter $\xi \in \mathbb{R}^d$ with law $\mathbb{P}$, constraint map g , and risk level $\varepsilon \in (0, 1)$, the individual and joint chance constraints read

$$\mathbb{P}(g(x, \xi) \leq 0) \geq 1 - \varepsilon, \quad \mathbb{P}(g_r(x, \xi) \leq 0, r = 1, \dots, m) \geq 1 - \varepsilon. \quad (15)$$

A third, equivalent form deserves its own display because the methodological chapters lean on it constantly. Writing the requirement in terms of the violation event,

$$\mathbb{P}(g(x, \xi) > 0) \leq \varepsilon, \quad (16)$$

makes the safety interpretation direct: the optimizer may move x anywhere it likes as long as the estimated probability of violation stays below the budget ε . The violation form is also the form in which sample-based methods naturally operate, since counting violations among samples estimates the left side of Eq. (16) directly. Throughout Part 3, the scalar random variable $Y(x) = g(x, \xi)$ introduced in Chapter 1 carries this event: $\{g(x, \xi) > 0\}$ is exactly $\{Y(x) > 0\}$, and every approximation method in this chapter is at bottom a different way of estimating the upper tail of $Y(x)$ at zero.

The pair in Eq. (15) conceals an asymmetry that practitioners learn quickly. Individual constraints control each requirement in isolation and say nothing about simultaneous satisfaction; the joint constraint controls the system but is strictly harder, both statistically (the event is an intersubsection) and computationally (the feasible set geometry worsens). The classical bridge is Boole’s inequality: enforcing individual constraints at levels ε_r with $\sum_r \varepsilon_r \leq \varepsilon$ implies the joint constraint at level ε , at a cost in conservatism that grows with m . The sharper treatment of joint constraints through a scalar violation statistic is deferred to Part 3, where it forms research question RQ4.

2.2.2 Convexity islands: the log-concavity theory

The single most important positive structural result in the field is due to Prékopa. A measure is log-concave when the logarithm of its density is concave, a class that includes the Gaussian, the uniform on a convex set, the exponential, and many other standard laws. Prékopa’s theorem states that if ξ has a log-concave distribution and the set $\{(x, \xi) : g(x, \xi) \leq 0\}$ is convex, then the probability function $p(x) = \mathbb{P}(g(x, \xi) \leq 0)$ is log-concave in x , and consequently every upper level set $\{x : p(x) \geq 1 - \varepsilon\}$ is convex (Prékopa, 1971, 1995). On this island, chance-constrained programming is convex programming: local solutions are global, duality holds, and mature solvers apply.

Figure 10 makes the theorem and its boundary visible in a computed example. For a Gaussian parameter and a bilinear constraint, the 90% level set of $p(x)$ is convex, panel (a). Replacing the Gaussian by a two-component mixture, a minimal departure from log-concavity, produces the non-convex level set of panel (b), the continuous analogue of the union-of-half-spaces construction in Chapter 1. The figure settles a question of attribution that matters for everything downstream: non-convexity is not caused by the constraint function, which is identical in both panels, but by the shape of the distribution. The geometry of the feasible set is inherited from the geometry of the law of ξ , and no amount of cleverness on the constraint side can repair what the distribution side has broken.

This inheritance is exactly why the log-concavity island, elegant as it is, carries so little of the practical load. The distributions documented in the application domains of Chapter 1, skewed and heavy-tailed forecast errors, multimodal disturbance mixtures, heavy-tailed returns, are precisely the laws that leave the log-concave class. The conclusion is unavoidable and worth stating without decoration: the positive theory applies where the uncertainty is well behaved, and the uncertainty that motivates

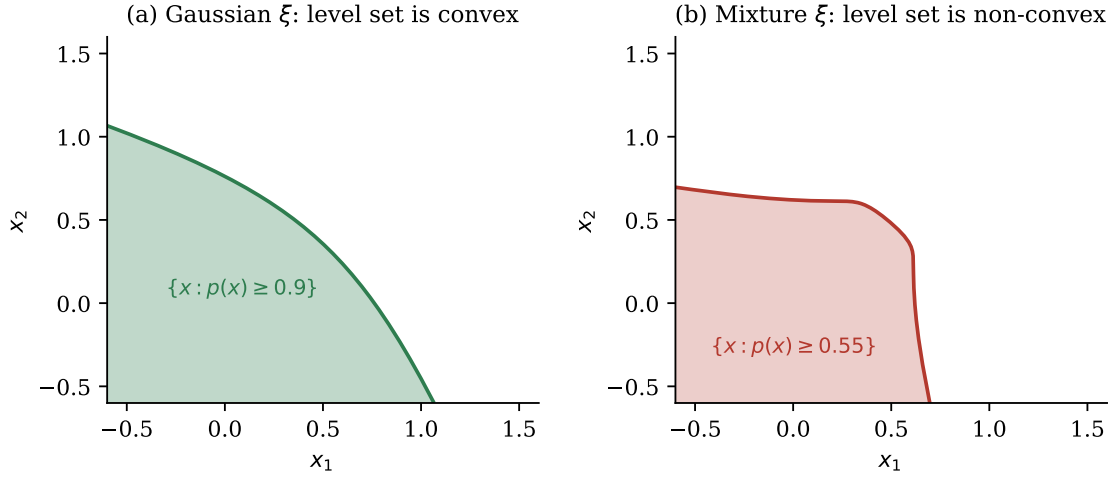


Figure 10: Prékopa’s theorem and its boundary, computed for the bilinear constraint $\xi^\top x \leq 1$ in $n = 2$. (a) For Gaussian ξ , a log-concave law, the probabilistic feasible set $\{x : p(x) \geq 0.9\}$ is convex, as the theory guarantees. (b) For a two-component Gaussian mixture, which is not log-concave, the level set $\{x : p(x) \geq 0.55\}$ develops the non-convex geometry anticipated by the discrete example of Chapter 1: the region bulges along two directions, one per mixture mode, and contains pairs of points whose midpoints are infeasible.

chance constraints in the first place rarely is. The general case is not an edge case. It is the case.

2.2.3 The general case

Outside the islands, the three difficulties of Chapter 1 stand in full force, and restating them in the violation-form language of Eq. (16) is more than housekeeping: the methods of the next subsection are organized by which difficulty each one attacks, and the restatement is the index of that organization. The feasible set $\{x : \mathbb{P}(Y(x) > 0) \leq \varepsilon\}$ is in general non-convex, so global optimality is out of reach and the realistic target is a high-quality local solution from a nonlinear programming method. The probability $\mathbb{P}(Y(x) > 0)$ has no closed form: it is an integral of an often unknown density over a region that moves with x , so it must be estimated, almost always from samples. And the natural sample estimate is an average of indicator functions, a piecewise-constant object whose gradient is zero almost everywhere, useless to exactly the solvers that handle nonlinearity well.

The landscape of Section 2.3 is best previewed as a matrix of concessions. Every classical method buys back tractability by surrendering one of three things: *generality* (the parametric route assumes the distribution), *exactness* (the convex restrictions enforce

a stronger constraint than required), or *smoothness* (the sample-enforcement routes hand the solver a non-differentiable or combinatorial problem). No classical method preserves all three; this is not an accident of history but a record of where six decades of effort have found the trade-offs to lie. The case this thesis builds, beginning in Section 2.4, is that estimating the density of $Y(x)$ with a kernel estimator concedes none of the three outright, paying instead with a bias that can be controlled and, in the biased-KDE construction, even directed to the safe side.

2.3 Standard Approximation Methods and Their Trade-offs

This subsection positions the thesis relative to the landscape of existing approaches, and it is built as a progression rather than a catalogue. Each method is presented in the same fixed rhythm, core idea, main advantage, primary limitation, and each method enters the survey as the field’s answer to the limitation of the one before it: the scenario approach removes the distributional assumption the Gaussian route cannot survive without, sample average approximation recovers the slack the scenario approach forfeits, the convex restrictions restore the tractability that sample averaging destroys, quantile smoothing recovers the exactness the restrictions surrender, and the blackbox approach strips away the last remaining assumption and pays the full price of doing so. The subsection closes with the comparison this ordering has been building toward, first as a table and then as a computed experiment on a problem where the truth is known exactly.

2.3.1 Parametric and Gaussian reformulations

The core idea of the parametric route is to assume the law of ξ and exploit it analytically. The Gaussian case with a constraint linear in ξ is the showpiece. For $g(x, \xi) = \xi^\top x - b$ with $\xi \sim \mathcal{N}(\mu, \Sigma)$, the residual $Y(x) = \xi^\top x - b$ is itself Gaussian with mean $\mu^\top x - b$ and variance $x^\top \Sigma x$, and the chance constraint collapses to an exact deterministic statement,

$$\mathbb{P}(\xi^\top x \leq b) \geq 1 - \varepsilon \iff \mu^\top x + \Phi^{-1}(1 - \varepsilon) \|\Sigma^{1/2} x\|_2 \leq b, \quad (17)$$

where Φ^{-1} is the standard normal quantile (Charnes u. Cooper, 1963; Shapiro u. a., 2014). For $\varepsilon \leq 1/2$ the right side of Eq. (17) is a second-order cone constraint: convex, smooth, and solvable at scale by every modern conic solver.

The advantage is total within its domain: exactness, convexity, smoothness, and speed, simultaneously. This is why the Gaussian reformulation dominates practice in fields such as chance-constrained optimal power flow (Bienstock u. a., 2014) and why it serves as the reference baseline of the experimental chapters of this thesis. When the Gaussian assumption holds, the problem identified in Chapter 1 simply does not exist.

The limitation is that the entire construction is the assumption. Every step of Eq. (17) flows from the normality of $Y(x)$, and Chapter 1 quantified what happens when that normality is false: a skewed law whose tail mass is underestimated by a factor of seven turns a certified 0.5% failure rate into an actual rate above 1.5%. The reformulation does not degrade gracefully; it reports a guarantee with full confidence whether or not the guarantee is true. Extensions to other named families and to Gaussian-like laws exist, including analytic gradient formulae for the probability function via spheric-radial decomposition (van Ackooij u. Henrion, 2014), but they relocate the assumption rather than remove it.

2.3.2 The scenario approach

If the Gaussian route fails because of its assumption, the natural response is to make none, and the scenario approach makes none with full commitment: it abandons the distribution entirely and enforces the constraint on data. Given independent samples $\xi_1, \dots, \xi_N$, the chance constraint is replaced by N hard deterministic constraints,

$$g(x, \xi_i) \leq 0, \quad i = 1, \dots, N. \quad (18)$$

The randomness now lives in the optimization problem itself, whose solution x_N^* is a random variable, and the theory asks how likely that solution is to satisfy the original chance constraint.

The advantage is the strength of the answer. For convex problems, Calafiore u. Campi (2006) showed that $N \geq \frac{2}{\varepsilon} \left(\ln \frac{1}{\beta} + n \right)$ samples suffice for x_N^* to satisfy the chance constraint at level ε with confidence $1 - \beta$, and Campi u. Garatti (2008) sharpened this to an exact, distribution-free feasibility certificate. These are finite-sample, a priori guarantees of a kind almost no other method offers, obtained without assuming anything about the law of ξ . The implementation is also disarmingly simple: sample, then solve a deterministic program.

The limitations are structural. The program grows with N , one hard constraint per sample, so precisely the regime where the statistics are comfortable is the regime

where the optimization is large. The approach cannot exploit the slack the chance constraint permits: it defends every sampled scenario, including the $\lfloor \varepsilon N \rfloor$ worst ones it is entitled to ignore, so it is conservative by construction and grows more conservative as N increases, a behavior the experiment of Section 2.3.7 displays vividly. And the guarantees are proven for convex problems; in the nonlinear optimal control settings of this thesis, they do not transfer.

2.3.3 Sample average approximation

The scenario approach is conservative because it defends samples the chance constraint entitles it to ignore. Sample average approximation is the correction: it keeps the samples but restores the slack, allowing exactly the violation budget the constraint grants. The violation probability in Eq. (16) is replaced by its empirical counterpart,

$$\hat{v}_N(x) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{g(x, \xi_i) > 0\} \leq \hat{\varepsilon}, \quad (19)$$

where the surrogate level $\hat{\varepsilon}$ may be set below ε to buy statistical confidence. The estimator is unbiased and the approach is statistically consistent: as $N \rightarrow \infty$, solutions of the approximated problem converge to solutions of the true one under mild conditions (Luedtke u. Ahmed, 2008; Pagnoncelli u. a., 2009). Unlike the scenario approach, SAA is allowed to discard its worst samples, so it tracks the true constraint rather than restricting it.

The advantage is exactly this statistical fidelity: Eq. (19) is the right quantity to compute, the empirical frequency of the event the constraint speaks about, with tunable conservatism through $\hat{\varepsilon}$ and well-developed asymptotic theory.

The limitation is that Eq. (19) is computationally hostile. The indicator makes $\hat{v}_N$ piecewise constant in x : its gradient vanishes almost everywhere and fails to exist on the boundary, so derivative-based nonlinear solvers receive no signal. The standard remedy is to encode the counting with binary variables, $g(x, \xi_i) \leq M z_i$, $\sum_i z_i \leq \lfloor \hat{\varepsilon} N \rfloor$, $z_i \in \{0, 1\}$, which is exact but converts the problem into a mixed-integer program whose difficulty grows combinatorially with N (Luedtke u. Ahmed, 2008). SAA thus poses a choice between a landscape with no gradients and a program with 2^N corners, and neither option fits the smooth optimal control pipelines this thesis targets.

2.3.4 CVaR and convex restrictions

SAA's failure is computational, not statistical, and the convex-restriction route responds on exactly that front: it trades the exact constraint for a stronger one whose enforcement is tractable, accepting conservatism as the price of convexity. Its central instrument is the conditional value-at-risk. For the residual $Y(x)$ and level ε , the Rockafellar-Uryasev representation

$$\text{CVaR}_{1-\varepsilon}(Y(x)) = \min_{t \in \mathbb{R}} \left\{ t + \frac{1}{\varepsilon} \mathbb{E}[(Y(x) - t)_+] \right\} \quad (20)$$

defines the average of the worst ε fraction of outcomes (Rockafellar u. Uryasev, 2000). Since the conditional value-at-risk dominates the value-at-risk, the implication $\text{CVaR}_{1-\varepsilon}(Y(x)) \leq 0 \Rightarrow \mathbb{P}(Y(x) > 0) \leq \varepsilon$ always holds: the CVaR constraint is an inner approximation of the chance constraint, and Nemirovski u. Shapiro (2006) established it as the tightest convex one in a natural family.

The advantages are convexity and safety in the same package. If $Y(x)$ is convex in x , then so is the CVaR constraint, the representation Eq. (20) is smooth after the standard reformulation of the positive part, and every feasible point of the restriction is provably feasible for the original chance constraint. Nothing in the construction requires knowing the distribution; the expectation in Eq. (20) is routinely replaced by a sample average, which preserves convexity.

The limitation is the direction of the inequality. CVaR controls the magnitude of violations, not merely their frequency, so it enforces more than the chance constraint asks: a decision with rare but large exceedances is rejected even when its violation probability is well inside the budget. The conservatism is largest exactly where chance constraints are most natural, under heavy-tailed residuals whose tail mean is far from the tail quantile. The experiment of Section 2.3.7 quantifies this gap on a skewed law: the CVaR solution overshoots the target reliability by a visible, structural margin that no amount of data removes.

2.3.5 Quantile smoothing

The survey has so far traded exactness for smoothness or smoothness for exactness; the question it forces is whether a method can refuse the trade. Quantile smoothing is the first demonstration that it can, and it is the closest methodological competitor to the approach of this thesis. Its starting point is the quantile equivalence. The chance

constraint holds exactly when the $(1 - \varepsilon)$ -quantile of the residual is nonpositive,

$$\mathbb{P}(Y(x) \leq 0) \geq 1 - \varepsilon \iff Q_{1-\varepsilon}(Y(x)) \leq 0, \quad (21)$$

and the empirical quantile, an order statistic of the residual samples, estimates the left side of Eq. (21) consistently. The empirical quantile is itself non-smooth in x , so Peña-Ordieres u. a. (2020) smooth it, constructing a differentiable sample-based approximation of $Q_{1-\varepsilon}(Y(x))$ and embedding it in a sequential quadratic programming framework with trust regions tailored to the resulting nonlinear program.

The advantages mirror those claimed for KDE in Chapter 1: distribution-free operation, smoothness, gradients that mainstream nonlinear solvers consume, and statistical guarantees for the smoothed estimator. The method demonstrates convincingly that the smooth-sample-approximation paradigm works at scale, and its existence is part of the evidence that this paradigm, rather than enforcement or restriction, is the right one for nonlinear problems.

The difference, and the limitation relative to this thesis, is the object being smoothed. Quantile smoothing approximates one functional of the residual law, the $(1 - \varepsilon)$ -quantile, and its smoothing is tied to that functional and that level. The KDE route estimates the residual density itself, from which the CDF at zero, its gradient, and quantiles at every level follow simultaneously in closed form; the same estimate also supports the biased construction of Part 3, which tilts the entire density surrogate to the safe side, a manipulation with no direct analogue at the level of a single quantile. The two methods are siblings, and the experimental chapters treat quantile smoothing as the fairest competitor.

2.3.6 The blackbox approach

Every method so far has exploited some structure: a distributional form, convexity, an analytic constraint map. The final method exploits none, and it closes the survey for two reasons that should be stated openly. It is where the assumption-stripping logic of the progression terminates, the limiting case in which nothing whatsoever is presumed about the problem. And it is where Part 3 begins, because it is the approach a practitioner with a simulator and no further structure would try first, so its precise mode of failure is the correct motivation for everything this thesis builds. The core idea is to treat the violation probability as a blackbox. The estimator $\hat{v}_N(x)$ of Eq. (19) can be evaluated at any x by running the simulator N times and counting failures;

nothing about g , the law of ξ , or the shape of the feasible set needs to be known. Optimization then proceeds with methods that consume only function values: penalty formulations driven by derivative-free solvers such as Nelder-Mead or pattern search, surrogate-assisted search, or grid-and-refine heuristics over low-dimensional decision spaces.

The advantage is universality. The blackbox approach applies to discontinuous simulators, legacy codes, and constraint maps with no analytic form, settings where every other method in this subsection is inapplicable in principle. It is also honest: $\hat{v}_N(x)$ is an unbiased estimate of the true violation probability, so the quantity being thresholded is statistically the right one, with none of the conservatism of scenarios or CVaR.

The limitations are what Part 3 is built to overcome, and they are worth stating sharply because they motivate everything that follows. The response surface seen by the optimizer is a staircase: piecewise constant in x , with zero gradient almost everywhere and jumps of size $1/N$ wherever a sample crosses the constraint boundary, all of it corrupted by sampling noise of order $N^{-1/2}$. Derivative-free methods tolerate this landscape but pay in evaluations, and their cost grows rapidly with the dimension of x ; in the optimal control problems of this thesis, where x is a discretized trajectory with hundreds of components, the approach is computationally out of the question. The blackbox is not wrong; it is blind. Chapter 3 develops it carefully, quantifies exactly where it fails, and uses that failure to motivate the smooth reformulation that the rest of the thesis studies.

2.3.7 Trade-offs in one view

The progression is complete, and its lesson can now be stated as a single claim the rest of the subsection substantiates: the classical methods partition the desirable properties among themselves, and none holds all of them at once. Table 1 condenses the survey into the four properties that matter for this thesis: whether the method is distribution-free, whether it hands a smooth problem to the solver, whether it is conservative, and what kind of feasibility guarantee it offers. The two KDE rows preview Part 3, and the distributionally robust row records the hybrid paradigm noted in Section 2.1.2, which wraps a Wasserstein ambiguity ball around the empirical law and obtains exact reformulations with finite-sample guarantees at the price of additional conservatism (Chen u. a., 2024).

Tables flatten behavior, so Fig. 11 runs the comparison as an experiment on the one-

Table 1: Approximation methods for chance constraints, compared on the four properties most relevant to nonlinear optimization and optimal control. “Smooth” means the reformulated problem presents differentiable constraints to the solver. Guarantees are classified as finite-sample (holding at fixed N with stated confidence), statistical (confidence statements for the estimator), or asymptotic (consistency as $N \rightarrow \infty$).

Method	Distribution-free	Smooth	Conservative	Guarantee
Gaussian reformulation	No	Yes	No	None if law wrong
Scenario approach	Yes	No	Moderate, grows in N	Finite-sample*
SAA (indicator)	Yes	No	Tunable via $\hat{\epsilon}$	Asymptotic
CVaR restriction	Partial	Yes	Yes, structural	Deterministic bound
Quantile smoothing	Yes	Yes	No	Statistical
Blackbox (DFO)	Yes	No	No	Asymptotic
KDE, unbiased	Yes	Yes	No	Asymptotic
KDE, biased	Yes	Yes	By construction	Asymptotic + safe side
Wasserstein DRO	Yes	Partial	Yes	Finite-sample

*For convex problems (Campi u. Garatti, 2008); does not transfer to the nonlinear setting.

dimensional dispatch problem of Chapter 1, where the lognormal demand law is known exactly and the true reliability of any decision can be computed in closed form. Five methods produce a capacity decision from N samples at target reliability 0.95, and the figure reports the reliability actually achieved, averaged over 400 independent sample draws per N . The behaviors predicted by the survey appear exactly. The Gaussian reformulation converges confidently to the wrong answer near 0.937, an unsafe plan certified as safe, and more data does not repair it because the error is in the assumption, not the estimate. The scenario approach climbs toward reliability one, drifting further from the target as N grows, the conservatism that defends samples it is allowed to ignore. CVaR stabilizes near 0.978, safe but structurally above target, paying for tail magnitudes the constraint never asked it to control. SAA and KDE both converge to the target, the defining behavior of methods that estimate the constraint rather than assume or restrict it; their curves are nearly indistinguishable here because the quantity is a scalar quantile, but only the KDE decision was produced by a smooth, differentiable construction, which is the property that matters once the decision variable stops being a scalar.

The experiment fixes the gap this thesis occupies. What is missing from the classical landscape is a method that is simultaneously distribution-free, smooth, non-conservative by default but conservative on demand, and equipped with convergence guarantees. The two KDE rows of Table 1 claim exactly this combination. The remainder of the chapter introduces the estimator behind the claim and the literature that

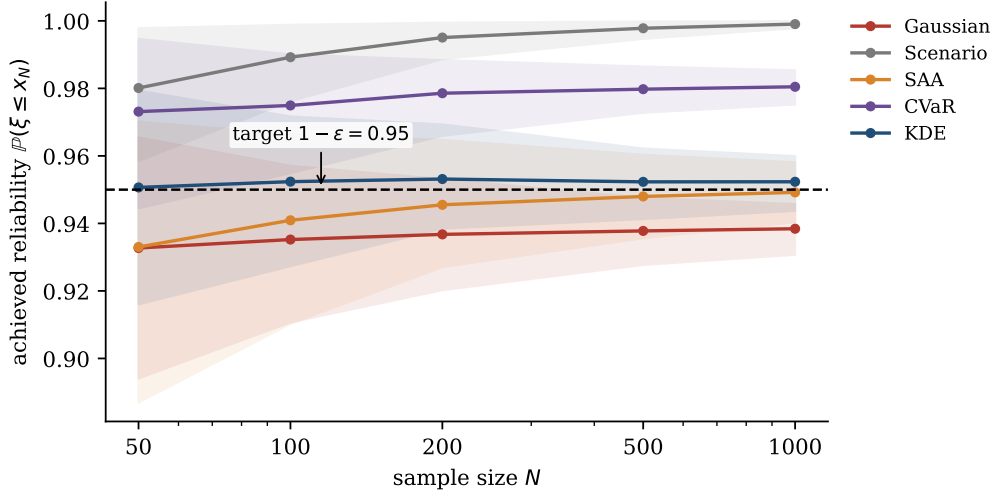


Figure 11: The survey of Section 2.3 as a computed experiment. Five methods choose a capacity from N samples of the lognormal demand of Chapter 1 at target reliability $1 - \varepsilon = 0.95$ (dashed); curves show the true reliability of the resulting decisions, averaged over 400 trials, with 10th-to-90th percentile bands. The Gaussian reformulation converges to an unsafe 0.937; the scenario approach and CVaR are persistently conservative; SAA and the KDE-smoothed quantile converge to the target. Only the KDE construction among the convergent pair is smooth in the decision variable.

has already tested it.

2.4 Kernel Density Estimation

The survey ended on a specification: a method that is distribution-free, smooth, non-conservative by default and conservative on demand, with convergence guarantees. The instrument that meets it is seventy years old. Kernel density estimation is standard equipment in statistics and largely unknown equipment in optimization, so this subsection introduces it deliberately, from the definition through the choices a user must make, with one organizing emphasis throughout: the property optimization cares about and statistics rarely examines, the behavior of the estimator as a function to be differentiated and embedded in a solver. Read with that emphasis, the classical theory of this subsection is not background; it is an inventory of exactly the assets and exactly the liabilities the reformulation of Part 3 inherits.

2.4.1 The Parzen-Rosenblatt estimator

Given scalar samples $y_1, \dots, y_N$ drawn from an unknown density ρ , the kernel density estimator of Rosenblatt (1956) and Parzen (1962) is

$$\hat{\rho}_h(y) = \frac{1}{Nh} \sum_{j=1}^N K\left(\frac{y - y_j}{h}\right), \quad (22)$$

where the kernel K is a nonnegative function integrating to one and the bandwidth $h > 0$ sets the smoothing scale. The mechanism is transparent: each sample is replaced by a small probability bump of width proportional to h , and the bumps are averaged. The estimator is itself a density, it inherits the smoothness of K , and its cumulative distribution function is the average of integrated kernels, $\hat{F}_h(y) = \frac{1}{N} \sum_j \mathcal{K}((y - y_j)/h)$ with $\mathcal{K}(t) = \int_{-\infty}^t K(s) ds$, available in closed form whenever $\mathcal{K}$ is. These are precisely the objects assembled in the residual-space construction of Chapter 1.

The estimator is consistent under remarkably weak conditions: if $h \rightarrow 0$ and $Nh \rightarrow \infty$ as $N \rightarrow \infty$, then $\hat{\rho}_h$ converges to ρ in mean integrated squared error for essentially any density of interest (Parzen, 1962; Devroye u. Györfi, 1985). No shape is assumed and none is imposed; given enough data the estimator recovers skewness, heavy tails, and multimodality alike. This distribution-free consistency is the property that earns KDE its place in this thesis, since the distributions of Chapter 1 are exactly the ones that defeat parametric fits.

For the optimization context, one more elementary observation carries disproportionate weight. The estimator Eq. (22) is a finite sum of shifted, scaled copies of a single fixed function. When the samples $y_j = g(x, \xi_j)$ depend on a decision variable, every analytic property of the estimate, its value, its integral, its derivative with respect to x , reduces by the chain rule to evaluations of K , $\mathcal{K}$, and the derivatives of g . There is no inner optimization, no numerical integration, and no nondifferentiable operation anywhere in the pipeline. The statistical literature prizes Eq. (22) for its flexibility; this thesis prizes it equally for being, computationally, a closed-form object.

2.4.2 Kernel choices

The kernel determines the local shape of the bumps, and three choices matter for this thesis. The Gaussian kernel $K(u) = (2\pi)^{-1/2} e^{-u^2/2}$ is smooth everywhere, has the standard normal CDF as its integrated kernel, and yields infinitely differentiable estimates

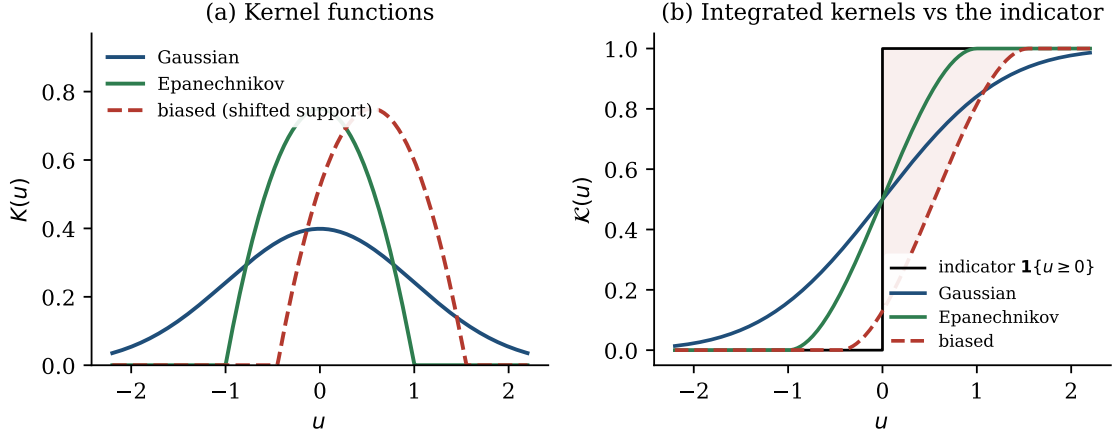


Figure 12: Kernels and their integrated forms. (a) The Gaussian and Epanechnikov kernels, both symmetric, and a biased kernel obtained by relocating the Epanechnikov support, the simplest illustration of the biasing idea formalized by the split-Bernstein construction of Keil u. a. (2021). (b) The corresponding integrated kernels against the indicator they replace. Symmetric kernels smooth the step from both sides and can overestimate the safe probability; the biased kernel’s integrated form sits on one side of the step (shaded gap), so the probability estimate it produces errs toward safety by construction.

with convenient gradients; it is the default choice throughout the unbiased developments of Part 3. The Epanechnikov kernel $K(u) = \frac{3}{4}(1 - u^2)_+$ is optimal for density estimation in the mean integrated squared error sense (Epanechnikov, 1969); its compact support also makes it the natural raw material for kernel modifications, since the entire kernel can be relocated. The split-Bernstein kernels of Keil u. a. (2021) are piecewise polynomial constructions engineered for a property neither classical kernel has: their integrated kernels bound the indicator function from the safe side, which is the seed of the biased, conservative reformulation developed in Part 3.

Figure 12 displays all three kernels and, in panel (b), the property that organizes Part 3. A symmetric kernel’s integrated form crosses the indicator at its jump: it underestimates the contribution of safely negative residuals and overestimates that of violating ones, and the net effect on the probability estimate can fall on either side. The biased kernel’s integrated form sits entirely on one side of the indicator, so every sample’s contribution errs in the same, safe direction. Anticipating the formal treatment, this single picture is the biased-KDE method: choose the kernel so that smoothing and conservatism are the same act.

Beyond these three, classical theory offers a reassurance that simplifies the rest of the thesis: the choice among standard symmetric kernels affects estimation efficiency only marginally, with the Gaussian sacrificing a few percent of efficiency relative to

the Epanechnikov optimum (Silverman, 1986). The kernel is therefore not where the estimation battle is won or lost. That battle belongs entirely to the bandwidth.

2.4.3 Bandwidth and the bias-variance trade-off

The bandwidth governs everything. Small h produces an estimate that follows the samples closely: low bias, high variance, and a ragged density whose spurious modes are artifacts of the particular draw. Large h produces a stable but blurred estimate: low variance, high bias, with genuine features such as modes and sharp tails smeared away. The mean integrated squared error makes the trade-off exact. For a symmetric kernel and a twice-differentiable density, the asymptotic expansion

$$\text{AMISE}(h) = \underbrace{\frac{R(K)}{Nh}}_{\text{variance}} + \underbrace{\frac{h^4 \mu_2(K)^2 R(\rho'')}{4}}_{\text{squared bias}}, \quad R(\phi) = \int \phi(s)^2 ds, \quad \mu_2(K) = \int s^2 K(s) ds, \quad (23)$$

is minimized at $h^* = [R(K)/(\mu_2(K)^2 R(\rho''))]^{1/5} N^{-1/5}$, giving the optimal rate $\text{AMISE}(h^*) = O(N^{-4/5})$ (Silverman, 1986). Figure 13 shows both faces of the trade-off computed on the bimodal residual law used throughout this thesis: panel (a) the visual pathology of under- and oversmoothing on a single sample, panel (b) the empirical bias-variance decomposition over 120 replications, with the V-shaped total error and its interior minimizer.

Since h^* depends on the unknown curvature $R(\rho'')$, practice relies on data-driven rules. Silverman’s rule of thumb, $h = 0.9 \min(\hat{\sigma}, \text{IQR}/1.34) N^{-1/5}$, calibrates the constant to a Gaussian reference and is robust enough to serve as the default in most software (Silverman, 1986). Scott’s rule, $h = \hat{\sigma} N^{-1/(d+4)}$ per dimension, generalizes the same idea to multivariate estimation (Scott, 1992). Cross-validation and plug-in methods estimate the curvature term from the data and adapt to non-Gaussian shapes at additional computational cost. For density estimation as an end in itself, any of these rules is adequate, and the near-optimal bandwidth of Fig. 13(a) is the value such rules produce on this sample.

A caveat must be placed on record here, because it grows into a research question later. Every rule above minimizes density estimation error, the AMISE of Eq. (23). Nothing in their derivation knows that the estimate will be thresholded at zero, differentiated with respect to a decision variable, and handed to a nonlinear solver as a constraint. In the optimization context the bandwidth shapes the feasible region and the gradient

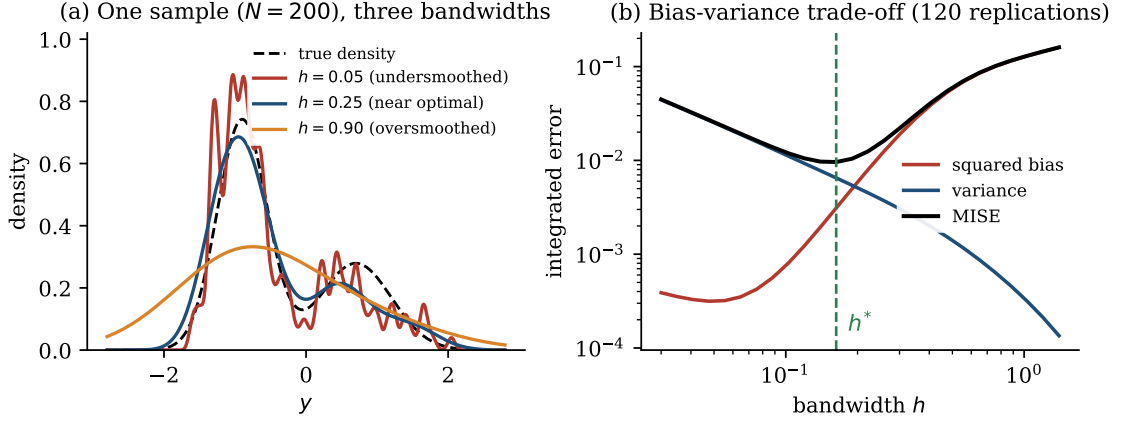


Figure 13: The bandwidth trade-off, computed on the bimodal law used in the residual-space illustration of Chapter 1. (a) One sample of size $N = 200$ estimated at three bandwidths: $h = 0.05$ invents spurious structure, $h = 0.90$ erases the two genuine modes, and $h = 0.25$ recovers the truth (dashed). (b) Empirical decomposition of the integrated error over 120 replications: variance falls and squared bias rises in h , and their sum is minimized at an interior h^* , here close to the $O(N^{-1/5})$ prediction of Eq. (23).

field, and the h that best reconstructs the density need not be the h that best locates the optimizer. This distinction, AMISE optimality versus optimization optimality, is the substance of research question RQ3 and is studied experimentally in Part 4.

2.4.4 Multivariate KDE and the curse of dimensionality

The estimator extends to d -dimensional samples by replacing the scalar bandwidth with a bandwidth matrix H , $\hat{\rho}_H(y) = \frac{1}{N} \sum_j |H|^{-1/2} K(H^{-1/2}(y - y_j))$, and the AMISE analysis extends with it. The conclusion, however, reverses sign. The optimal bandwidth scales as $O(N^{-1/(d+4)})$ and the optimal error as $O(N^{-4/(d+4)})$ (Silverman, 1986; Scott, 1992): the convergence rate decays with every added dimension, and the sample size required to hold accuracy constant grows exponentially in d . Figure 14 translates the rate into numbers. Holding the accuracy achievable with $N = 100$ samples in one dimension fixed, five dimensions already demand on the order of 10^4 samples and ten dimensions on the order of 10^6 . This is the curse of dimensionality in its sharpest form, and it is the fundamental limitation of multivariate kernel estimation.

The reader has already seen the answer this thesis gives. The residual-space construction of Chapter 1 never asks KDE to work in the dimension of ξ : an individual chance constraint produces a scalar residual whatever the dimension of the uncertainty, and the estimation problem is one-dimensional by design, where the $O(N^{-4/5})$

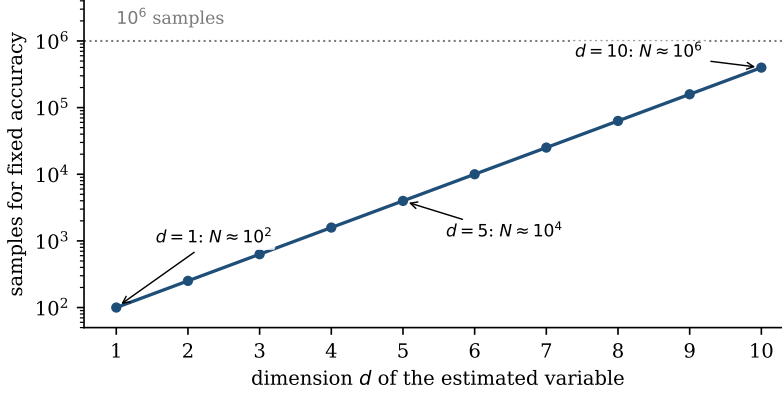


Figure 14: The curse of dimensionality for kernel density estimation: sample size required to maintain the accuracy achievable with $N = 100$ in one dimension, under the AMISE rate $O(N^{-4/(d+4)})$. The required sample size grows exponentially in d , reaching roughly 10^6 at $d = 10$. This figure is the quantitative case for estimating the scalar constraint residual rather than the uncertainty vector itself.

rate is available and a few hundred samples suffice. The curse retreats to one clearly marked location, joint chance constraints, whose residual vector has dimension m . Keeping that dimension out of the estimator, by collapsing the residual vector through a scalar violation statistic, is precisely the extension pursued as research question RQ4, and Fig. 14 is the reason it is worth pursuing.

2.5 KDE in the Chance-Constrained Literature: A Literature Map

The combination of kernel density estimation and chance constraints is no longer a proposal awaiting validation; it is a literature with a decade of results across four application domains and, since 2025, a convergence theory. What that literature has never received is an organizing narrative, and supplying one is not a courtesy to the reader but a load-bearing step of the argument: only when the papers are read as a single development does it become visible that the engineering line and the theoretical line have never been connected, which is the gap this thesis fills. The narrative has three movements, displayed as a timeline in Fig. 15: independent invention, the safety turn, and the arrival of theory.

2.5.1 First constructions: process systems and aerospace

The data-driven construction enters the literature with Calfa u. a. (2015) in process systems engineering. Facing production planning problems whose right-hand-side un-

certainty was known only through historical data, the authors replaced the unknown distribution with a kernel-smoothed estimate and derived deterministic reformulations for both individual and joint chance constraints, demonstrating them on industrial planning and scheduling cases. The paper established the template every later work follows, samples to smoothed estimate to deterministic constraint, and demonstrated that the resulting problems are solvable with standard nonlinear programming machinery.

Caillaud u. a. (2018) carried the construction into optimal control, the setting of this thesis. Treating launcher ascent and the Goddard problem with uncertain parameters, the authors reformulated the chance constraints through kernel estimates and solved the resulting smooth optimal control problems with indirect and direct methods, establishing feasibility of the approach in exactly the trajectory-optimization context where the scenario and SAA routes are least comfortable. The aerospace line and the process systems line were developed independently, and their convergence on the same construction from different needs is early evidence that the construction is natural rather than contrived.

In parallel, the energy systems community arrived by a third route. Ciftci (2017) and Ciftci u. a. (2019) built adaptive-KDE chance constraints for microgrid energy management under renewable uncertainty, and the dispatch line that followed (Wu u. a., 2021; Wu u. Kargarian, 2023) pushed the construction to multivariate kernels for joint constraints in power-system scheduling. Three communities, three motivations, one estimator: by the end of this first movement the empirical case was broad, but every guarantee in it was inherited from generic KDE consistency, with nothing said about the optimization problems built on top.

2.5.2 The safety line: biased kernel density estimation

The first movement shares an unexamined weakness: a symmetric kernel smooths in both directions, so the estimated safe probability can exceed the true one, and a solution certified at level $1 - \varepsilon$ can be infeasible for the original constraint. The contribution of Keil u. a. (2020) and Keil u. a. (2021) is to make the smoothing itself safe. Their split-Bernstein kernels are constructed so that the integrated kernel bounds the indicator from the conservative side, as illustrated in Fig. 12(b); the resulting probability estimate is biased low by design, and feasibility for the approximated constraint implies feasibility for the true one up to the estimator's statistical error. The method is demonstrated on chance-constrained soft lunar landing solved by direct collocation,

the benchmark this thesis adopts.

The follow-up work of Keil u. Rao (2021) addresses the computational cost that conservatism introduces, proposing a warm-start strategy that solves a sequence of problems with decreasing bandwidth and switching kernels, using each solution to initialize the next. The pair of papers constitutes the primary algorithmic reference of this thesis: between them they supply the safe reformulation, the kernel constructions, and the practical solution strategy that Part 3 develops, dissects, and extends.

2.5.3 Networks, breadth, and the arrival of theory

The construction's breadth test came from infrastructure. Schuster u. a. (2022) applied KDE-reformulated probabilistic constraints to stationary gas networks and dynamic water contamination networks, derived the gradients of the kernel-estimated probability analytically, and used them in necessary optimality conditions, comparing the KDE route against spheric-radial decomposition, the analytic alternative available under Gaussian-type laws (van Ackooij u. Henrion, 2014). Adjacent developments multiplied the evidence: nonparametric prediction intervals for wind power (Wan u. a., 2020), Bayesian nonparametric gas-power flow (Wang u. a., 2021), KDE-based tail-risk portfolio optimization (Liu u. a., 2022), KDE-driven distributionally robust load shedding (Qi u. a., 2026), and Hilbert-space embeddings of nonparametric chance constraints in robotics (Gopalakrishnan u. a., 2018).

What every entry above still lacked was a statement about the optimization problem itself: KDE consistency guarantees that the estimated probability converges, not that the minimizer of the estimated problem converges to the minimizer of the true one. Schuster (2025) supplies exactly this missing piece, proving convergence of optimal values and solution sets of the KDE-approximated chance-constrained problem to those of the original as $N \rightarrow \infty$, under verifiable conditions on the kernel, the bandwidth sequence, and the problem data. With this result the literature acquires the shape visible in Fig. 15: a decade of engineering constructions, and a convergence theory developed separately that certifies the class they belong to.

The gap that remains is the connection itself, and it is the gap this thesis occupies. The engineering line and the theoretical line have not been brought into systematic contact: no work verifies the convergence conditions for the biased constructions used in practice, no controlled study quantifies how bandwidth choices made by AMISE rules behave inside an optimizer, and no unified experimental comparison places the KDE

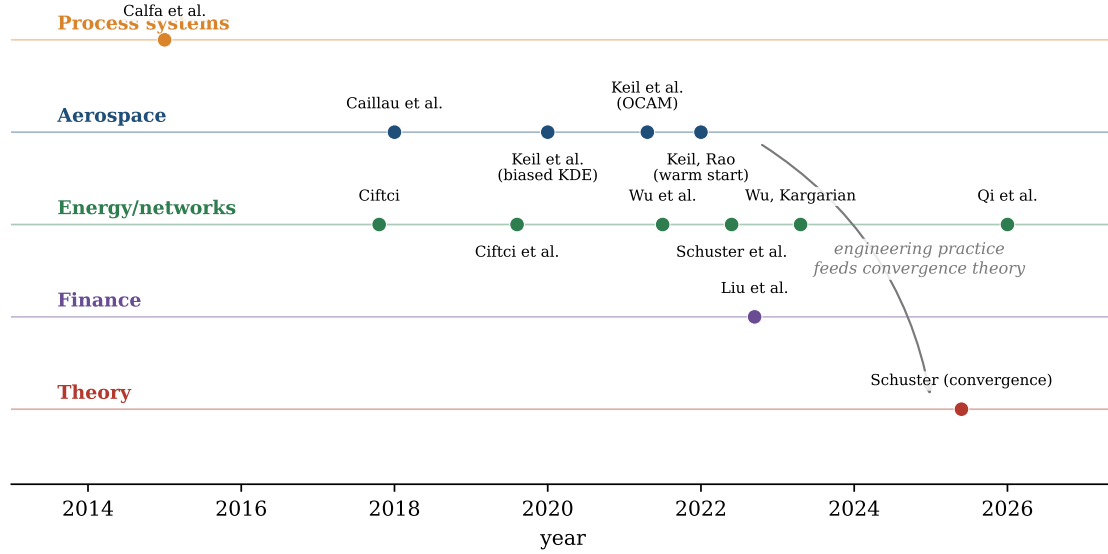


Figure 15: The KDE chance-constraint literature as a timeline, organized by application stream. The construction appears independently in process systems (Calfa u. a., 2015), aerospace (Caillau u. a., 2018), and energy systems (Ciftci, 2017; Ciftci u. a., 2019); the biased-KDE safety line (Keil u. a., 2020, 2021; Keil u. Rao, 2021) and the network applications (Schuster u. a., 2022) deepen it; and the convergence theory of Schuster (2025) arrives a decade after the first engineering constructions, closing the loop the gray arrow traces.

family against the classical methods of Section 2.3 across sample sizes, risk levels, and problem types, including the optimal control setting. Building that bridge, formulation by formulation in Part 3 and experiment by experiment in Part 4, is the work the next chapter begins, starting where any practitioner would: with the blackbox.

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Erklärung

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